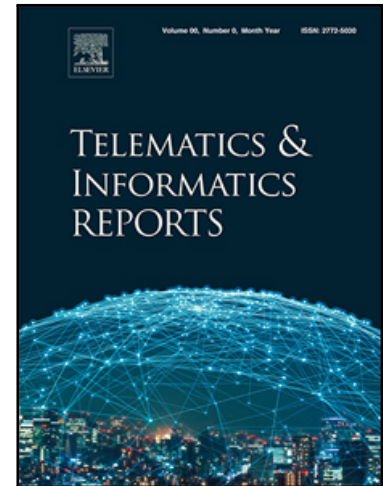


How AI Tutor Features Influence Motivation and Learning Outcomes:
Psychological Pathways in High School STEM Education

Thanh Chung Pham , Thi Thuy An Ngo

PII: S2772-5030(26)00055-1
DOI: <https://doi.org/10.1016/j.teler.2026.100344>
Reference: TELER 100344



To appear in: *Telematics and Informatics Reports*

Received date: 10 December 2025
Revised date: 6 May 2026
Accepted date: 11 May 2026

Please cite this article as: Thanh Chung Pham , Thi Thuy An Ngo , How AI Tutor Features Influence Motivation and Learning Outcomes: Psychological Pathways in High School STEM Education, *Telematics and Informatics Reports* (2026), doi: <https://doi.org/10.1016/j.teler.2026.100344>

This is a PDF of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability. This version will undergo additional copyediting, typesetting and review before it is published in its final form. As such, this version is no longer the Accepted Manuscript, but it is not yet the definitive Version of Record; we are providing this early version to give early visibility of the article. Please note that Elsevier's sharing policy for the Published Journal Article applies to this version, see: <https://www.elsevier.com/about/policies-and-standards/sharing#4-published-journal-article>. Please also note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2026 Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license
(<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

HIGHLIGHTS

- A multidimensional S-O-R framework explains how AI tutor features shape STEM learning outcomes.
- Control increases learning satisfaction but did not significantly influence motivation.
- Efficiency, enjoyment, content, personalization, and security enhances both motivation and satisfaction.
- Both learning motivation and satisfaction significantly predicts learning outcomes.
- Motivation has a stronger impact on learning outcomes than satisfaction.

How AI Tutor Features Influence Motivation and Learning Outcomes: Psychological Pathways in High School STEM Education

Thanh Chung Pham¹, Thi Thuy An Ngo^{2*}

¹Department of International Business, FPT University, Can Tho, Vietnam. Email: ChungPTCS170363@fpt.edu.vn

^{2*}Corresponding Author. Department of Admissions, FPT University, Can Tho, Vietnam. Email: AnNTT24@fe.edu.vn

Contact: FPT University, 600 Nguyen Van Cu Street, An Binh Ward, Can Tho City, Vietnam.

Abstract

AI tutors have emerged as a critical technological intervention to meet the escalating demands for personalized and accessible STEM education. Guided by the Stimulus–Organism–Response (S-O-R) framework, this study investigates the influence of AI tutors on high school students' learning processes by examining how six key stimuli, including control, efficiency, enjoyment, learning content, personalization, and security, affect the organismic states of learning motivation and satisfaction, which subsequently influence learning outcomes. Using Partial Least Squares Structural Equation Modeling (PLS-SEM) on data collected from 478 Vietnamese high school students, the results reveal differentiated effects across system dimensions. Control operates as a hygiene factor, defined as a baseline functional requirement that prevents dissatisfaction but does not inherently drive engagement, which enhances learning satisfaction but does not significantly predict motivation. In contrast, efficiency, enjoyment, learning content, personalization, and security serve as dual drivers that positively influence both motivation and satisfaction. Critically, both learning motivation and satisfaction significantly contribute to improved learning outcomes. Theoretically, the study advances AI-in-education research by validating a multidimensional AI tutor framework tailored to the K–12 STEM context. Practically, the findings highlight that while hygiene factors like intuitive interfaces and reliable performance reduce user frustration, meaningful engagement depends on adaptive learning pathways, emotionally supportive interactions, and robust data protection. These insights provide actionable guidance for developers and educators seeking to enhance the pedagogical effectiveness and responsible deployment of AI tutors in high school STEM education.

Keywords: AI tutor, STEM education, learning motivation, learning satisfaction, learning outcomes, high school students

1. Introduction

The global education landscape is currently undergoing a significant and complex transformation driven by two concurrent developments: the recognized importance of Science, Technology, Engineering, and Mathematics (STEM) education and the rapid advancement of Artificial Intelligence (AI) (Yang et al., 2025). STEM education at the high school level, particularly among students aged 15 to 17, is increasingly acknowledged as a crucial foundation for developing the future workforce (Imeri et al., 2024). This specific educational stage is essential for forming the scientific knowledge, critical thinking skills, and academic attitudes that shape future career pursuits. Simultaneously, AI has emerged as a transformative force with the potential to address significant educational challenges and fundamentally alter traditional teaching and learning paradigms (Marengo & Pange, 2024). The convergence of these trends is most evident in the development of Intelligent Tutoring Systems (ITS) or AI tutors, which provide adaptive learning pathways, real-time feedback, and individualized support (Adel, 2024; Gomes, 2025). While these tools are widely considered a promising means of addressing global demand for high-quality STEM instruction (Villegas-Ch et al., 2025), scholars caution that technological sophistication alone does not guarantee improved learning outcomes (Bulathwela et al., 2024). Current evidence remains mixed and inconclusive regarding AI tutors' pedagogical effectiveness (Ouyang et al., 2022), underscoring the need for further empirical validation.

In Vietnam, STEM education has become a national priority, positioned as a strategic driver of modernization and economic growth (Nguyen, 2024). This policy environment represents a “technological leapfrogging” strategy, where AI is framed not merely as a supplementary tool but as a critical infrastructure to bypass traditional resource deficits (Hang et al., 2024). Despite strong governmental commitment, a significant gap persists between national policy ambitions and actual classroom realities, where STEM implementation is often constrained by the pedagogical legacy of Confucian-heritage culture, which traditionally prioritizes teacher-centered authority and rote learning (Tran, 2024). Consequently, AI-powered Educational Technologies (EdTech) are being rapidly adopted as a mechanism to bridge the gap between policy intentions and classroom practice (Le et al., 2025). Vietnamese high school students, digitally fluent members of Generation Z, exhibit high receptiveness to emerging technologies (Trinh et al., 2024), engaging with AI through a unique sociotechnical lens that balances collective social expectations with individual achievement goals (Nguyen et al., 2023). However, how these learners actually interact with AI tutors, and how such interactions influence their psychological states and learning performance, remains insufficiently explored. Vietnam’s intense academic pressure, coupled with a transition toward student-centered learning within a historically hierarchical system, creates a unique psychological tension in AI adoption that may diverge from Western experiences. Thus, understanding this demographic provides both local relevance and broader insight into AI adoption within rapidly evolving, resource-constrained, yet culturally distinct education systems.

Existing literature in the international context has explored various determinants of AI tutor effectiveness. Previous studies have indicated that features such as learner control allow students to regulate their pacing and improve engagement (Brusilovsky, 2024), while personalization capabilities have been linked to higher relevance and attention (Ahmed & Aziz, 2025). Similarly, the quality of learning content (Nalluri et al., 2024) and the perceived efficiency of the system (Farhan et al., 2024) have been identified as predictors of system usage. Furthermore, the element of enjoyment (Fan & Zhang, 2024) and concerns regarding security (Panditharathna et al., 2024) are known to impact user trust and adoption. The significance of these attributes is further underscored by recent pilot surveys investigating student perceptions, which suggest that the successful integration of AI-enhanced systems relies heavily on how learners evaluate these specific adaptive functionalities (Gyonyoru & Katona, 2024). While these studies have illuminated individual components of AI-based learning, most investigations have been conducted in higher education settings (Nagaraj et al., 2023; Sun et al., 2025). As a result, the K–12 sector, and particularly high school students, remains significantly understudied (Zhang & Tur, 2024). This gap is consequential because high school students possess fundamentally different motivational structures, cognitive needs, and levels of learner autonomy compared to the adult learners found at universities or colleges (Yaban & Gaschler, 2025). Moreover, existing research frequently treats AI tutors as a single, uniform construct (Sofologi et al., 2025), limiting the ability to identify which specific features drive positive learning experiences (Liu C. et al., 2025). Few studies decompose AI tutors into multidimensional components and examine their distinct effects on learner psychology and outcomes in real classroom contexts (Lin et al., 2025).

To address this limitation, the current study conceptualizes AI tutors as a multidimensional construct comprising six distinct dimensions: control, efficiency, enjoyment, learning content, personalization, and security. These components were derived through a rigorous inductive approach, systematically synthesizing recurring attributes from prominent AI in education research to ensure a comprehensive representation of the technological environment. This decomposition is supported by a comprehensive overview of the diverse concepts and applications of AI-based adaptive learning, which highlights the necessity of addressing multiple system facets to achieve pedagogical goals (Gyönyörű & Katona, 2025). This six-dimensional framework is not ad hoc but is grounded in the necessity to capture the functional, pedagogical, and ethical spheres of human-AI interaction (Pham & Ngo, 2026). Specifically, control and personalization represent the adaptive nature of AI that fosters learner agency (Adayilo et al., 2026), while efficiency and learning content address the core utility and instructional quality essential for academic tasks (Sajja et al., 2026). Furthermore, enjoyment and security encompass the affective and trust-based dimensions that are often overlooked in purely cognitive models but are vital for adolescent users (Li et al., 2024; Khoso et al., 2025). By integrating these six factors, the model provides a theoretically exhaustive view of the AI tutor, moving beyond monolithic descriptions to identify the specific psychological pathways triggered by distinct system attributes.

In the specific context of educational research within Vietnam, there is a scarcity of empirical studies that systematically link multidimensional AI attributes to psychological outcomes in STEM learning. While general studies on EdTech adoption in Vietnam exist, they often focus on infrastructural barriers or general acceptance rates rather than the psychological mechanisms of learning. The literature suggests that psychological constructs such as learning motivation (Nguyen & Nguyen, 2025; Vu et al., 2022) and learning satisfaction (Pham et al., 2025; Tran et al., 2024) are critical drivers of academic success. However, it remains unclear how specific AI tutor attributes, such as efficiency, control, or security, independently shape these internal states for Vietnamese adolescents. A fully integrated model linking multidimensional AI tutor attributes with both psychological mechanisms and learning outcomes remains largely absent from current scholarship (Duong et al., 2025). These limitations reveal two major gaps: (1) a demographic gap concerning the absence of K–12-focused AI tutor research and (2) a theoretical gap involving the lack of a mechanistic

explanation to describe how specific technological attributes influence students' internal states and academic results.

Addressing these gaps, the present study aims to examine how specific attributes of AI tutors influence students' psychological states and, subsequently, their STEM learning outcomes using the Stimulus–Organism–Response (S-O-R) framework (Mehrabian & Russell, 1974). The S-O-R model is adopted because it provides a structured explanation of how external system features (Stimulus) shape learners' internal cognitive and emotional states (Organism), which then lead to observable educational results (Response). In this study, the Stimulus is operationalized through six distinct dimensions of AI tutors: control, efficiency, enjoyment, learning content, personalization, and security. The Organism consists of two key psychological constructs, learning motivation and learning satisfaction, while the Response is defined as students' STEM learning outcomes. Accordingly, the study pursues three core research objectives: (1) to determine the extent to which each AI tutor attribute predicts learning motivation and learning satisfaction; (2) to assess how these psychological states influence students' learning outcomes; and (3) to develop and empirically validate an integrated S-O-R model that explains the mechanism through which AI tutors contribute to STEM learning performance in Vietnamese high schools. By analyzing these relationships, the study offers a detailed and mechanism-based understanding of how interactions with AI tutors shape learner experiences and academic results.

This research aims to establish a novel theoretical framework by expanding the S-O-R model to capture the complex interplay between AI system design and psychological states in high school STEM education. Moving beyond previous general applications of technology-mediated learning, the study is designed to demonstrate the empirical validity of a specific configuration of six AI tutor stimuli, including control, efficiency, enjoyment, learning content, personalization, and security, in driving academic success. The theoretical novelty will reside in the granular operationalization of how these technical attributes interact to simultaneously trigger dual psychological organism factors: learning motivation and learning satisfaction. By seeking to confirm these parallel pathways, the research intends to provide precise evidence-based bridges between specific system configurations and student outcomes, filling a critical gap in the existing literature. Crucially, the study is expected to translate these prospective findings into high-impact practical strategies, offering actionable blueprints for developers and educators. From a design perspective, the anticipated results will likely mandate a transition from surface-level personalization to dynamic “gap-detection algorithms” and “personalized mastery dashboards” that visualize learning trajectories in real time. Furthermore, by identifying security as a potential driver of satisfaction, the research aims to highlight the necessity of transparent “privacy shields” and “permission-based sharing” to foster psychological safety. For educational practitioners, the model is expected to validate the integration of AI tutors as diagnostic “early-warning systems,” allowing teachers to use AI-generated analytics for timely human intervention. By grounding these projected recommendations in verified S-O-R pathways, this study seeks to ensure that future technological implementations in STEM are strategically engineered to optimize the affective and cognitive states necessary for measurable pedagogical improvement.

2. Literature review

2.1. Theoretical background

2.1.1. AI tutor

AI tutors, commonly referred to as AI tutors, represent a class of educational technologies designed to provide automated instructional support, adaptive guidance, and real-time learning assistance to students. Conceptually, AI tutors operate as intelligent systems capable of delivering pedagogically meaningful scaffolding that aligns with the cognitive and affective needs of high school learners (Melo, 2024). These systems are programmed to interpret student inputs, diagnose misconceptions, and provide targeted feedback, and they frequently emulate the instructional functions traditionally associated with human tutors. For example, AI tutors can guide students step by step through complex physics problems, generate personalized mathematics practice sets, or facilitate virtual laboratory simulations that mirror authentic scientific inquiry (Anoir et al., 2024). This functional design situates AI tutors within the broader context of information systems research, where system quality, information quality, and service quality have long been used to evaluate technological performance. In high school STEM contexts, the focus extends to how effectively AI tutors support both cognitive mastery and emotional engagement (Abreh et al., 2025). In this study, the term “AI tutors” refers to systems that integrate technological and pedagogical components to enhance students' understanding and motivation in STEM subjects (Shi et al., 2024).

The growing integration of AI tutors in high school education highlights their expanding role in contemporary STEM classrooms. As digital learning ecosystems evolve, AI-driven adaptive instruction has become an important catalyst for improving personalization and instructional precision, particularly for diverse cohorts of adolescents aged 15 to 17 (Abisoye, 2024). Students in this age group are digital natives who expect interactive, dynamic learning environments rather than static digital materials. They increasingly depend on AI tutors to receive real-time homework support,

visualize abstract scientific concepts through simulations, and receive guidance tailored to their individual learning pace (Aggarwal et al., 2024). These expectations underscore the role of AI tutors as central tools for enhancing accessibility, responsiveness, and engagement in STEM learning. Their continued adoption depends heavily on students' perceptions of how well the systems meet their learning needs, a point that is especially important because learning motivation and learning satisfaction are theorized in this study as key mechanisms linking AI tutor features to STEM learning outcomes (Jeilani & Abubakar, 2025).

Guided by the proposed theoretical framework (Figure 1), this study conceptualizes AI tutors through six distinct features that are particularly relevant to the STEM learning experiences of high school students. The first feature is control, which refers to students' perceived ability to regulate their learning activities, such as selecting appropriate task difficulty levels or navigating among STEM modules at their own pace (Qin et al., 2025). This sense of control promotes autonomy, although it must be supported by appropriate scaffolding to prevent cognitive overload. The second feature is efficiency, which concerns the extent to which the AI tutor helps students manage study time effectively by offering immediate explanations or hints while they confront complex STEM tasks. Efficiency must, however, avoid encouraging superficial shortcuts that undermine deeper reasoning processes (Kestin et al., 2025). The third feature is enjoyment, which reflects the degree to which students perceive their learning experience as intrinsically engaging or stimulating. Enjoyment in STEM is often fostered through gamification, interactive visualizations, and hands-on virtual experimentation that sustain interest in demanding subjects like physics or chemistry (Wang H. et al., 2021). The fourth feature is learning content, which focuses on the accuracy, rigor, and curricular alignment of instructional materials delivered by the AI tutor. This feature is especially critical in STEM learning because factual precision is essential, and inaccurate content generation may lead to significant conceptual errors (Parviz, 2024). The fifth feature is personalization, which captures the system's ability to tailor learning pathways to the individual needs and foundational gaps of each student. Personalization supports competency development in cumulative fields such as mathematics by adjusting the instructional sequence according to learners' evolving performance (Kayyali, 2025). The sixth feature is security, which involves the protection of sensitive data for minor students. Strong privacy safeguards form the basis of trust among students, parents, and educators and are essential for the large-scale adoption of AI tutors in school settings (Walker et al., 2023).

By examining these six AI tutor features, this study provides a comprehensive understanding of how the specific attributes of AI tutors shape learning motivation, learning satisfaction, and ultimately learning outcomes in high school STEM education. This multidimensional perspective allows for a more nuanced analysis of the psychological mechanisms through which AI tutors influence both cognitive and affective dimensions of student learning.

2.1.2. Learning outcomes

Learning outcomes are conceptualized as the measurable knowledge, skills, and competencies acquired by students through specific instructional processes (Goss, 2022). In the landscape of high school STEM education, these outcomes are multifaceted, encompassing objective academic gains, such as scientific conceptual depth and mathematical problem-solving, as well as subjective developments like self-efficacy in technical domains (Rasyid et al., 2023). Within the S-O-R framework, learning outcomes represent the response to environmental stimuli. While traditional instructional models often treat these outcomes as direct consequences of pedagogical input, contemporary scholarship suggests that the effectiveness of AI tutors is not a foregone conclusion. Indeed, a significant debate exists regarding whether digital interventions consistently outperform human-led instruction. While some assessments demonstrate significant progress through AI-mediated mastery (Deng et al., 2025; Wongmahesak et al., 2025), other studies have highlighted a no significant difference phenomenon, where the introduction of advanced technology fails to yield superior academic dividends compared to conventional methods (Wei, 2025; Yang et al., 2025). This tension suggests that learning outcomes are not merely a product of technology usage but are contingent upon how system features interact with the learner's internal state.

The relationship between AI tutor features (stimulus) and learning success is characterized by both promise and pedagogical risk. Established literature emphasizes that dimensions such as personalization, efficiency, and learning content are primary drivers of performance. For instance, adaptive sequencing can optimize the acquisition of complex STEM concepts by matching content to individual proficiency (Liu J. et al., 2025), while efficient system design mitigates cognitive load during demanding calculus or physics tasks (Ngu & Phan, 2024). However, a critical synthesis of recent research reveals that these benefits are often unevenly distributed. Critics of high-level personalization argue that over-tailored content may inadvertently create "filter bubbles" or "knowledge silos," where students are shielded from the productive struggle necessary for deep conceptual change (Kausar & Kumar 2025). Furthermore, while interactive features are associated with higher performance (Melo, 2024), empirical evidence from certain STEM contexts suggests that excessive system-led control can lead to "cognitive passivity," where students rely on AI prompts rather than developing independent logical reasoning (Zhai et al., 2025). Consequently, the impact of AI features on

learning outcomes must be viewed as a delicate balance between supportive scaffolding and the potential for over-dependence, a theoretical nuance that explains the mixed effects reported in broader educational technology research.

Within the proposed S-O-R conceptual model, the pathway from AI features to learning outcomes is mediated by the organism, the internal psychological states of learning motivation and learning satisfaction. This sequential mechanism is grounded in expectancy-value theory, where motivation acts as the catalyst for deep engagement with scientific materials (Arsyad et al., 2024). Highly motivated learners are generally more inclined to persist through the rigorous demands of STEM curricula, which theoretically translates into improved academic performance (Gyasi et al., 2021). Parallel to this, satisfaction serves as a critical affective filter; positive associations have been found between student satisfaction and ultimate academic success (Bernardo et al., 2025; Oliso et al., 2024). Yet, this psychological pathway is not always linear. Theoretical tensions arise when considering that high system efficiency might lead to satisfaction but not necessarily to the “desirable difficulties” required for long-term retention (Klein & Klein, 2025). If an AI tutor prioritizes immediate satisfaction through ease of use, it may fail to trigger the deep cognitive processing required for robust learning outcomes.

Accordingly, this study posits a more nuanced sequential mechanism: high-quality AI tutor dimensions, such as personalization, efficiency, and content, do not influence learning outcomes in a vacuum. Rather, they must be calibrated to foster internal states of motivation and satisfaction that are conducive to rigorous academic effort. By acknowledging that these features can occasionally result in null or contradictory effects if poorly aligned with the learner’s psychological needs, this research provides a theoretically mature exploration of how AI tutors contribute to, or potentially hinder, academic improvement among high school STEM learners.

2.1.3. The Stimulus-Organism-Response (S-O-R) theory

The Stimulus-Organism-Response (S-O-R) paradigm, originally conceptualized within the field of environmental psychology by Mehrabian and Russell (1974), provides a robust theoretical foundation for explaining how environmental cues influence human internal states and subsequent behaviors. This classic framework posits a linear yet dynamic process wherein external environmental stimuli (S) act as primary triggers that affect the internal cognitive and emotional states of an organism (O), which collectively drive specific behavioral responses (R). In its initial conception, the model was primarily utilized to understand physical retail environments, analyzing how sensory inputs such as lighting or layout impacted consumer emotions (Erensoy et al., 2024). However, scholars have significantly expanded its scope over the decades to encompass diverse disciplines, ranging from organizational behavior to information systems and virtual environments (Zhao et al., 2020). While the model is frequently praised for its structural clarity, critics often argue that its application in digital contexts can overlook the non-linear complexity of human-computer interaction, where stimuli do not always yield predictable internal shifts (Mikić et al., 2025). The organism component is particularly critical as it represents the internal processing mechanism, often comprising affective and cognitive evaluations, which serves as the intervening psychological bridge between the external world and the final action (Zhang et al., 2021). Furthermore, contemporary adaptations of the framework have validated its efficacy in predicting user reactions to digital interfaces, suggesting that the stimulus is no longer confined to physical space but includes technological features and digital atmospherics (Kexin & Teo, 2023). Nevertheless, a critical tension remains in S-O-R research regarding whether technological stimuli serve as empowering tools or cognitive burdens, a debate that is particularly salient in the evolving landscape of AI (Bansal et al., 2025). Consequently, the S-O-R model remains a predominant lens for dissecting the causal chain from environmental inputs to behavioral outputs across both physical and virtual contexts (Wu et al., 2025).

In the realm of educational technology and digital learning systems, the S-O-R framework has been extensively applied to decipher the complex interactions between technological affordances and learner behaviors. However, the existing literature presents a dichotomy; while many studies emphasize the positive reinforcement of psychological states, others highlight a “technological paradox” where advanced features can lead to cognitive overload or reduced agency (Abbas et al., 2024). Numerous studies have utilized this lens to investigate how specific features of e-learning platforms serve as stimuli that shape students’ psychological engagement and learning habits. For instance, Dai et al. (2024) utilized the framework to demonstrate that interactive features in online classrooms significantly bolster students’ flow experience, thereby encouraging continuous usage intentions. Similarly, regarding AI in education, Alwakid et al. (2025) argued that the perceived intelligence and anthropomorphism of AI agents enhance learner trust and emotional connection, which in turn fosters sustained student-agent interaction. Conversely, Johnpaul et al. (2025) cautioned that an over-reliance on AI “intelligence” may inadvertently diminish a student’s own metacognitive development, suggesting that the “stimulus” of AI does not universally result in positive “organismic” growth. Research by Yang (2024) further highlighted that system quality and information clarity in mobile learning apps act as foundational stimuli driving learner satisfaction, ultimately leading to higher mobile learning adoption rates. In a related context, Cheng (2024) established that gamification elements within educational software effectively trigger intrinsic motivation, subsequently resulting in

higher task completion rates. Yet, Ruijia et al. (2025) found that excessive gamification can occasionally distract learners from core content, leading to “null effects” on actual learning outcomes despite high initial motivation. Moreover, Jiang et al. (2024) observed that the visual appeal and navigational ease of learning management systems influence students’ cognitive absorption, which translates into improved learning performance. Finally, Pan et al. (2024) provided empirical evidence suggesting that personalized feedback mechanisms in digital tutors positively impact students’ perceived value, subsequently fostering active participation behaviors. Despite these supportive findings, unresolved debates persist regarding whether high levels of “control” and “personalization” actually overwhelm high school learners, who may lack the self-regulatory skills to manage such autonomy (Jauhiainen & Guerra, 2024). These scholarly works collectively underscore the versatility of the S-O-R model while simultaneously revealing a need for a more nuanced exploration of when and why these stimuli succeed or fail.

Drawing upon this established theoretical lineage, the present study adapts the S-O-R framework to scrutinize the efficacy of AI tutors within the specific context of high school STEM education. Rather than assuming a guaranteed positive progression from technology to performance, this study acknowledges the inherent tensions between AI features and student psychology. As illustrated in the research model, the stimulus component comprises six distinct AI tutor factors: control, efficiency, enjoyment, learning content, personalization, and security. These elements are conceptualized as the environmental attributes of the AI-driven learning ecosystem that students encounter directly. The inclusion of “security” and “control” as stimuli specifically addresses the unresolved debate concerning user anxiety and data privacy in AI education, factors that can negatively disrupt the internal state of the learner. These external technological stimuli are posited to influence the organism, which represents the students’ internal psychological processing and evaluation of their learning experience. Specifically, the organismic state is defined by two pivotal constructs: learning motivation and learning satisfaction. These internal states are not viewed as passive reactions but as complex cognitive and affective filters that can be either enhanced or suppressed by the AI environment. Finally, the response is operationalized as learning outcomes, representing the tangible academic results and performance achievements derived from the interaction. By employing this structured approach, the study posits that high-quality attributes of the AI tutor do not impact grades directly but rather shape the students’ internal motivational and satisfaction levels, which subsequently determine their academic success in STEM subjects. This application of the S-O-R model allows for a granular analysis that moves beyond descriptive summaries to explore how the specific, and sometimes contradictory, functionalities of AI technologies translate into tangible educational improvements through the psychological enhancement of the learner.

2.2. Hypothesis development

2.2.1. Impacts of control on learning motivation and learning satisfaction

In the context of AI tutors for STEM learning, control refers to the learner’s volitional agency and freedom in directing the learning process (Balo & Sanchez, 2025). This construct reflects students’ capacity to navigate, sequence, and personalize their interactions with the system in accordance with their own goals, thereby functioning as a mechanism that supports the psychological need for autonomy (Osei & Bjorklund, 2024). From a theoretical standpoint, autonomy is widely recognized as a fundamental driver of intrinsic motivation, particularly among adolescent learners. However, the relationship between control and learning motivation in AI-mediated environments is not straightforward. While autonomy-supportive perspectives suggest that greater control enhances learners’ intrinsic drive (Ruiz & Yabut, 2024), emerging evidence points to a “paradox of choice,” where excessive control may overwhelm learners and reduce engagement. For instance, McKay et al. (2025) reported null effects of control on motivation, suggesting that without sufficient prior knowledge, students may experience choice overload, leading to disengagement rather than empowerment. This tension suggests that the effectiveness of control depends on learners’ capacity to manage it. Recent studies have attempted to reconcile these contrasting findings by highlighting the role of self-regulatory competence and perceived self-efficacy. When learners possess sufficient capability to direct their learning process, control can foster intrinsic interest and persistence (Bognár et al., 2024). Similarly, allowing students to regulate their learning pace has been shown to enhance their motivation to master complex content (Xu et al., 2024). Furthermore, Guo et al. (2025) demonstrated that managing learning sequences can encourage perseverance, particularly when control aligns with students’ confidence in their abilities. Taken together, these findings suggest that control operates not as a universally beneficial factor, but as a conditional stimulus whose positive motivational effects are more likely to emerge under supportive conditions. Building on this integrative perspective, this study posits that, within appropriately designed AI tutoring systems, control features can effectively stimulate learners’ intrinsic motivation toward STEM learning. Accordingly, the following hypothesis is proposed:

- **H₁:** Control has a significant positive influence on learning motivation.

The influence of control on learning satisfaction reflects a similar pattern of theoretical complexity. On one hand, control

enhances satisfaction by enabling learners to act as active agents in the learning process, thereby increasing perceived personalization and reducing frustration (Liang et al., 2024; Gkintoni et al., 2025). When students are able to adjust learning paths or difficulty levels, they tend to perceive the system as more responsive to their needs, which leads to more positive evaluations (Tan et al., 2025). In addition, a sense of ownership over learning progress has been identified as a strong predictor of satisfaction (Rizki & Suprpto, 2024). On the other hand, excessive control may introduce cognitive burdens associated with decision-making. Tsabary and Hewitt (2025) found that highly open-ended AI environments can reduce satisfaction due to decision fatigue, particularly when learners are required to make frequent or complex choices. This suggests that the benefits of control may diminish when the system imposes high cognitive demands or lacks sufficient guidance. When interfaces are intuitive and provide adequate scaffolding, control enhances learners' sense of empowerment without inducing navigation anxiety (Bhardwaj et al., 2025). Under such conditions, control contributes to a more personalized and satisfying learning experience. Rizki and Suprpto (2024) also confirmed that ownership over progress is a strong predictor of satisfaction, it is essential to view control not as a guaranteed benefit but as a stimulus whose success is contingent upon the balance between student agency and system support. By synthesizing these conflicting viewpoints, this study posits that the autonomy provided by control features remains a fundamental driver of a positive educational experience. Based on this empirical grounding, the following hypothesis is proposed:

- **H₂:** Control has a significant positive influence on learning satisfaction.

2.2.2. Impacts of efficiency on learning motivation and learning satisfaction

Efficiency is defined as the AI tutor's operational performance and functional reliability in task execution (Maia et al., 2024). It reflects the system's ability to streamline interactions, deliver timely responses, and allocate instructional resources in a way that minimizes extraneous cognitive load. As such, efficiency primarily triggers learners' cognitive appraisal of instrumental utility and interaction fluency, enabling more focused, goal-directed learning processes (Moors, 2025). From a cognitive perspective, prior research generally supports the view that efficient systems enhance learning motivation by freeing up mental resources for deeper engagement with academic content. When interfaces reduce unnecessary effort, learners are better able to concentrate on mastering complex STEM concepts (Gavrilas & Kotsis, 2025; Gkintoni et al., 2025). In addition, adaptive efficiency such as personalized pathways and immediate feedback, has been shown to sustain motivation by maintaining an optimal balance between challenge and attainability (Huang et al., 2025; Tavdgiridze et al., 2024). This process reinforces learners' sense of competence, which is a critical driver of intrinsic motivation. Consistently, when students perceive that their time and effort are utilized productively, their willingness to persist in demanding tasks increases (Chisunum & Nwadiokwu, 2024), and the experience of smooth, uninterrupted progress further strengthens self-efficacy and long-term engagement (Bejtac, 2024). However, the positive effect of efficiency is not unequivocal. Emerging studies highlight a critical tension between efficiency and meaningful cognitive engagement. Excessive efficiency, particularly when driven by over-automation, may reduce opportunities for active processing, leading to superficial learning and diminished intrinsic motivation (Klein & Klein, 2025). Similarly, Mohamed et al. (2025) reported non-significant effects of efficiency on motivation when automated support resulted in student passivity. These findings suggest that efficiency may undermine motivation if it removes the "desirable difficulties" necessary for deep learning. To reconcile these perspectives, recent literature suggests that the motivational impact of efficiency depends on the extent to which it balances cognitive support with active learner involvement. When designed appropriately, efficient systems reduce unnecessary cognitive burden without eliminating meaningful challenge, thereby creating conditions that are conducive to sustained motivation. Building on this integrative view, this study posits that efficiency, as a core functional attribute of AI tutors, is more likely to enhance learning motivation when it supports rather than replaces cognitive effort. Accordingly, the following hypothesis is proposed:

- **H₃:** Efficiency has a significant positive influence on learning motivation.

The effect of efficiency on learning satisfaction follows a similar pattern of convergence and tension. Within technology acceptance research, efficiency is often closely associated with perceived ease of use, which serves as a fundamental antecedent of user satisfaction (Zhou et al., 2022). In AI-supported STEM learning contexts, systems that operate smoothly and minimize technical complexity tend to generate positive affective responses, thereby enhancing overall satisfaction (Liu & Liu, 2025). Pandita and Kiran (2023) argued that satisfaction is not solely a result of academic outcomes but is significantly influenced by the quality of the interaction; they maintain that an efficient AI tutor must provide a dependable interface to minimize technical frustration. Moreover, given that digital-native students typically hold high expectations regarding system responsiveness, efficient performance plays a crucial role in shaping their perceptions of comfort and usability (Reid et al., 2023). Nevertheless, recent studies indicate that efficiency alone may not guarantee a satisfying learning experience. Overly streamlined or automated interactions can lead to "algorithmic fatigue," where the system is perceived as mechanical and lacking human-like nuance (Yang et al., 2024). Furthermore, some learners derive satisfaction from the challenge inherent in problem-solving, suggesting that excessive ease may

reduce the perceived meaningfulness of the learning experience (Deepshikha, 2026). These contrasting findings highlight that satisfaction is influenced not only by functional performance but also by the qualitative nature of the interaction. Synthesizing these perspectives, this study conceptualizes efficiency as a critical determinant of learning satisfaction, particularly when supported by thoughtful system design that preserves learner involvement. Therefore, the following hypothesis is proposed:

- **H4:** Efficiency has a significant positive influence on learning satisfaction.

2.2.3. Impacts of enjoyment on learning motivation and learning satisfaction

In the context of AI-supported STEM learning, enjoyment is conceptualized not merely as entertainment but as a meaningful positive emotional state that arises when learners successfully engage with optimally challenging tasks (Mierzwa-Kamińska, 2025). While existing literature provides evidence that this emotional state acts as a driver of learning motivation, the relationship is not universally linear and remains a subject of intense debate. For instance, Lin-Siegler et al. (2023) showed that enjoyment influences students' behavioral responses when confronting challenging material, while Kassenkhan et al. (2025) argued that personalized learning environments can transform abstract problem-solving into intrinsically rewarding experiences. In addition, positive emotions have been found to broaden cognitive resources, thereby facilitating deeper engagement and sustained effort (Bustamante-Mora et al., 2025). These findings collectively position enjoyment as a key emotional driver of motivation in AI-mediated learning contexts. However, competing evidence highlights important limitations. Some studies point to a "novelty effect," whereby initial enjoyment diminishes over time and fails to sustain long-term engagement (Wu & Li, 2025). Moreover, excessive emphasis on enjoyment, particularly through gamified or entertainment-oriented features, may increase extraneous cognitive load or shift learners' focus toward superficial interaction rather than genuine mastery (Dunne, 2025; Rana & Chicone, 2025). This suggests that enjoyment can function as a double-edged construct, potentially undermining motivation if it is not aligned with pedagogical objectives. To reconcile these perspectives, recent literature emphasizes the importance of balancing enjoyment with cognitive challenge and instructional rigor. When learners experience enjoyment within non-judgmental and supportive environments, it reinforces intrinsic engagement without compromising learning depth (Akhter, 2025). Under such conditions, enjoyment enhances rather than distracts from goal-directed behavior, ultimately strengthening motivation. Consistent with this view, Ortiz-Rojas et al. (2025) confirmed that enjoyable learning experiences remain positively associated with heightened motivation in STEM contexts. Building on this integrative perspective, this study posits that enjoyment, when appropriately embedded within challenging and meaningful learning activities, serves as a critical emotional driver of motivation. Accordingly, the following hypothesis is proposed:

- **H5:** Enjoyment has a significant positive influence on learning motivation.

Beyond its motivational role, enjoyment is also widely recognized as a key antecedent of learning satisfaction, defined as learners' overall affective evaluation of their educational experience (Martin & Bolliger, 2022). Prior research consistently indicates that when learners perceive the learning process as enjoyable rather than merely functional, they are more likely to report higher satisfaction (Zardari et al., 2021). In AI-supported environments, features that evoke enjoyment such as adaptive pathways or engaging interactions, have been shown to contribute to more positive experiential evaluations (Yang et al., 2025). Despite this general consensus, the relationship between enjoyment and satisfaction is not without tension. Some studies suggest that high levels of enjoyment may not translate into satisfaction if learners perceive the content as lacking intellectual depth or rigor (Le, 2025). Similarly, while technically efficient systems can produce low satisfaction when perceived as monotonous (Guo et al., 2025), overly playful or entertainment-driven designs may frustrate learners who prioritize efficiency and task completion (Tan et al., 2023). These findings point to a fundamental trade-off between hedonic appeal and utilitarian value in shaping satisfaction. Further complexity arises from the dynamic interplay between challenge and affect. The "frustration-enjoyment" cycle observed in AI-supported STEM learning suggests that satisfaction often emerges not from constant enjoyment, but from the successful resolution of difficulty (Li et al., 2024). This implies that enjoyment contributes to satisfaction most effectively when it is intertwined with meaningful cognitive engagement rather than detached from it. Therefore, by synthesizing these conflicting perspectives, it is argued that enjoyment remains a primary lens for evaluation when it facilitates successful learning. Based on this empirical foundation, the following hypothesis is proposed:

- **H6:** Enjoyment has a significant positive influence on learning satisfaction.

2.2.4. Impacts of learning content on learning motivation and learning satisfaction

Learning content constitutes the core instructional component of AI tutoring systems, comprising STEM topics and interactive materials (Sun et al., 2025). While traditional perspectives assume a linear increase in engagement with higher content volume, recent debates suggest a far more complex relationship. For instance, AI-enabled content incorporating

simulations and dynamic representations has been shown to foster deeper engagement with complex concepts (Pellas, 2025). Similarly, the diversity and structured presentation of learning materials can reduce monotony and sustain persistence over time (Bansal et al., 2023; Wolf & Ebner, 2025). In addition, adaptive content that calibrates difficulty levels is often associated with maintaining learners in an optimal challenge zone, thereby preventing both boredom and discouragement (Gligorea et al., 2023). However, these positive effects are not universal. A growing body of research highlights that content richness may become counterproductive when it exceeds learners' processing capacity. In particular, highly interactive or information-dense materials can induce cognitive overload, especially among students with limited prior knowledge (Balaskas et al., 2025). Furthermore, adaptive mechanisms do not always yield the intended outcomes; when algorithmic adjustments fail to align with learners' actual self-regulation capabilities, their motivational impact may be negligible (Sharma et al., 2024). Additionally, overly structured or prescriptive content may restrict learners' sense of autonomy, potentially triggering psychological reactance and reducing intrinsic motivation. To reconcile these perspectives, recent studies suggest that learning content operates as a conditional regulatory mechanism rather than an inherently positive driver. Its motivational effectiveness depends on achieving a balance between intellectual challenge and cognitive accessibility, ensuring that content remains engaging without becoming overwhelming (Gkintoni et al., 2025). When this balance is maintained, content design can effectively support sustained motivation in AI-mediated STEM learning environments. Building on this integrative view, this study posits that appropriately designed learning content is likely to enhance students' motivation. Accordingly, the following hypothesis is proposed:

- **H7:** Learning content has a significant positive influence on learning motivation.

The impact of learning content on satisfaction reflects a similarly nuanced pattern. As the primary medium through which instruction is delivered, content quality serves as a central criterion for evaluating the overall learning experience. Prior research suggests that clearly organized and logically structured content reduces cognitive friction and contributes to more positive user evaluations (Gavrilas & Kotsis, 2025). Moreover, high-quality materials characterized by accuracy, sufficiency, and clarity are consistently associated with perceptions of effectiveness and increased satisfaction (Khairuddin et al., 2024; Ho et al., 2021). In contexts where instructional materials constitute the core service, content quality becomes a particularly salient determinant of user satisfaction (Fu et al., 2024). Nevertheless, the literature reveals several important tensions. Organizational clarity alone may not guarantee satisfaction if the content does not align with learners' preferences or learning styles (Balarezo et al., 2024). In addition, the so-called "technological paradox" suggests that advanced or feature-rich content may inadvertently distract learners from core learning objectives, thereby diminishing satisfaction (Wan & Zhao, 2024). Furthermore, even highly accurate content may fail to meet learners' expectations if it lacks contextual or cultural relevance. Another concern relates to information density, as excessive content, even when well-structured, can lead to "information fatigue," negatively affecting learners' emotional responses (Zhong & Gu, 2024). By recognizing these contradictions, this study moves away from the assumption of theoretical inevitability, arguing instead that satisfaction arises from a delicate balance between content depth and usability. This nuanced perspective acknowledges that while superior content is essential, its role in generating satisfaction is subject to the learner's internal processing limits. Based on this empirical foundation, the following hypothesis is proposed:

- **H8:** Learning content has a significant positive influence on learning satisfaction.

2.2.5. Impacts of personalization on learning motivation and learning satisfaction

In the context of AI tutoring for STEM, personalization represents the alignment of recommended content with the learner's specific interests, situational needs, and unique academic profile (Huang et al., 2025). It involves the system's dynamic adjustment of instructional pathways, content difficulty, and learning pace to match individual student characteristics. Theoretically, this construct fosters an affective sense of relevance and individual recognition, ensuring that the learning experience is perceived as personally significant and tailored to the learner's unique identity (Ding & Xue, 2025). However, the scholarly discourse remains divided regarding the magnitude of this effect. While traditional perspectives suggest personalization is a primary driver of motivation by accommodating varied levels of prior knowledge (Wong & Wong, 2021), contemporary critics argue that excessive tailoring may inadvertently diminish student agency or create a filter bubble that limits exposure to diverse problem-solving strategies. Ali and Elkot (2025) demonstrated that when students progress at their own speed and tackle problems that are appropriately challenging, they develop a profound sense of autonomy and competence that fuels intrinsic motivation. Similarly, personalized approaches can mitigate frustration and boredom commonly associated with standardized curricula, particularly in complex STEM domains (Zhai & Krajcik, 2024). These findings suggest that personalization enables learners to engage more deeply by making content more accessible and relevant. However, the effectiveness of personalization is not without contention. Emerging research highlights a "personalization paradox," wherein excessive or poorly implemented tailoring may reduce learner agency or constrain exposure to diverse problem-solving strategies. For example, Shin

(2023) observed that overly algorithmic personalization can lead to “algorithmic fatigue,” reducing sustained engagement. Moreover, Rad (2026) reported non-significant effects in certain STEM contexts, indicating that personalization alone may not be sufficient to overcome deeply rooted learning barriers such as math anxiety. These findings suggest that personalization does not universally translate into higher motivation and may even produce unintended negative effects when misaligned with learners’ needs. When personalization enhances relevance while preserving learners’ sense of control and cognitive challenge, it can effectively stimulate intrinsic motivation. Conversely, when it becomes overly prescriptive or restrictive, its benefits may diminish. Consistent with this view, Aunzo (2025) argued that personalization enhances motivation when it supports learners through challenging tasks without undermining their active involvement. Based on this empirical foundation, the following hypothesis is proposed:

- **H₉:** Personalization has a significant positive influence on learning motivation.

Personalization also serves as a critical antecedent of learning satisfaction by fundamentally shaping the perceived quality and relevance of the high school student’s experience. Prior research suggests that when AI systems tailor explanations and instructional content to individual comprehension levels, they create smoother and more effective learning interactions, thereby enhancing satisfaction (Tan et al., 2025). In addition, personalized feedback, particularly when it targets specific errors, has been shown to improve learners’ perceptions of support and reduce frustration in STEM learning contexts (Villegas-Ch et al., 2025). These findings highlight the importance of responsiveness and precision in fostering positive evaluations of AI-supported learning. Nevertheless, similar to its effect on motivation, the relationship between personalization and satisfaction is characterized by notable tensions. The personalization paradox suggests that highly tailored systems may raise concerns related to data privacy or the accuracy of learner profiling, potentially reducing satisfaction (Du et al., 2025). Furthermore, some learners may perceive overly personalized recommendations as restrictive or misaligned with their actual goals, leading to dissatisfaction. In certain cases, standardized or comparative feedback may even be preferred, as it provides clearer benchmarks for academic performance (Pellas, 2026). Despite these inconsistencies, a substantial body of research indicates that personalization generally enhances satisfaction when it is perceived as supportive rather than intrusive. For example, individualized feedback that validates learners’ efforts and provides clear guidance has been associated with higher satisfaction levels (Maier & Klotz, 2022). Moreover, when learners perceive that an AI system effectively understands and accommodates their specific challenges, their overall evaluation of the learning experience improves significantly (Glick et al., 2024). Synthesizing these perspectives, this study conceptualizes personalization as a key determinant of learning satisfaction, whose positive effects emerge when tailored support is balanced with user autonomy and perceived appropriateness. Therefore, the following hypothesis is proposed:

- **H₁₀:** Personalization has a significant positive influence on learning satisfaction.

2.2.6. Impacts of security on learning motivation and learning satisfaction

Security, defined as the rigorous protection of a learner’s personal identifiers and STEM data from unauthorized access, acts as a prerequisite for sustained learning motivation within AI educational environments (Al-Zahrani & Alasmari, 2024). However, the literature presents a significant tension regarding the privacy paradox, where students express high levels of concern yet maintain usage habits regardless of perceived risk. While some scholars argue that security is a primary driver, others, such as Micatka (2025), have found null effects where security features did not significantly alter long-term engagement levels. Tang et al. (2025) emphasized that a secure digital learning space fosters a sense of psychological safety that is critical for adolescents who rely on iterative trial-and-error processes to master complex scientific concepts. Motivation is not merely a product of protection but a nuanced outcome of the cognitive freedom that security affords. Popescu et al. (2024) indicated that when young learners perceive their sensitive data is effectively shielded through transparent policies, they are relieved of privacy-related concerns that otherwise consume cognitive resources. This reduction in cognitive load is not universally accepted; Bouchard (2025) suggested that for tech-savvy students, these concerns are often secondary to tool utility, creating a theoretical debate over the hierarchy of learner needs. Reduced risk enables them to focus attention fully on demanding learning tasks, which Al-kfairy et al. (2025) argued encourages them to experiment with novel problem-solving strategies without fear of judgment. Consequently, this safe environment helps maintain the intrinsic drive requisite for academic improvement. Conversely, Habbal et al. (2024) demonstrated that an environment perceived as lacking adequate security inhibits motivation because a fundamental lack of trust leads to reluctance to utilize the tool. Building on this integrative view, this study posits that security, as a core system attribute, positively influences learning motivation by fostering a safe and trustworthy learning environment. Accordingly, the following hypothesis is proposed:

- **H₁₁:** Security has a significant positive influence on learning motivation.

Security also plays a definitive role in shaping high school learners’ satisfaction with AI tutoring systems. According to

Jo & Park (2023), user satisfaction is strongly driven by system quality, which inherently includes the reliability, stability, and effectiveness of data protection mechanisms. In online STEM learning contexts, robust security mechanisms can signal professionalism and trustworthiness, thereby enhancing users' evaluations of the platform (Terzieva et al., 2025). Furthermore, when learners perceive that a system prioritizes the protection of their academic data, it strengthens perceived value and credibility, leading to more favorable experiences (Ogunmokun et al., 2021; Rohles et al., 2022). Nevertheless, empirical findings reveal several important tensions. Some studies report non-significant relationships between security and satisfaction, suggesting that improvements in security features do not always translate into higher user evaluations (Estipona & Santos, 2025). Additionally, the effectiveness of security in shaping satisfaction depends on its visibility; while clearly communicated security measures can enhance user confidence, invisible or background protections may have limited perceptual impact (Acheampong et al., 2025). Further complexity arises from the potential negative effects of excessive security measures. For example, overly restrictive authentication procedures may lead to "security fatigue," reducing satisfaction by increasing user effort and disrupting interaction flow (Al-Husari et al., 2025). Conversely, perceived weaknesses in data protection or lack of transparency can generate feelings of vulnerability, significantly diminishing satisfaction regardless of the system's pedagogical quality (Habbal et al., 2024). Taken together, these findings suggest that security influences satisfaction through a balance between protection and usability. When implemented effectively, providing visible assurance without imposing excessive burden, security enhances learners' confidence and contributes to a more positive overall experience. Based on these findings, this study conceptualizes security as an important determinant of learning satisfaction, whose positive effects emerge when protection mechanisms are both reliable and user-friendly. Therefore, the following hypothesis is proposed:

- **H₁₂:** Security has a significant positive influence on learning satisfaction.

2.2.7. Impacts of learning motivation on learning outcomes

Learning motivation functions as the essential internal psychological force that initiates and sustains goal-oriented educational activities, representing a crucial prerequisite for high school students attempting to master the cognitively demanding domain of STEM learning (Jannini et al., 2024). While prevailing research suggests a positive link, recent literature reveals significant tensions regarding the stability of this relationship. Within the specific context of AI-mediated environments, Chiu et al. (2024) provided evidence that motivation directly impacts outcomes by determining the magnitude of cognitive effort learners invest. Nevertheless, scholars such as Paul (2025) contended that high motivation can occasionally lead to cognitive "over-persistence," where students become frustrated by complex tasks rather than achieving success. Yu and Tao (2025) established a clear causal link between psychological drive and achievement, showing that students with elevated motivation levels not only actively utilize AI-provided resources but also employ deeper information processing strategies that facilitate substantial knowledge retention. This perspective is challenged by findings suggesting that even motivated students may struggle with knowledge transfer if the AI's feedback is insufficient (Lin & Chen, 2024). Furthermore, Duterte et al. (2024) confirmed that such motivational factors bolster academic performance metrics by fostering sustained interaction with the subject matter, which effectively reduces dropout rates. This impact is critically reinforced by Le et al. (2025), who observed that in autonomous AI-tutor settings, internal motivation acts as a more decisive determinant of performance than even the specific learning strategies employed by the student. By acknowledging these conflicting perspectives, it becomes evident that while motivation is necessary, its impact is contingent upon the alignment between learner drive and system support. Based on this empirical foundation, the following hypothesis is proposed:

- **H₁₃:** Learning motivation has a significant positive influence on learning outcomes.

2.2.8. Impacts of learning satisfaction on learning outcomes

Learning satisfaction is defined as a learner's subjective and cumulative evaluation of their experiences with AI tutors, reflecting the perceived value derived from interactions within educational environments (Seo & Um, 2023). However, the relationship between satisfaction and learning outcomes remains a subject of scholarly debate, as some contemporary studies report null or inconsistent effects where high levels of user satisfaction do not necessarily translate into objective academic gains (Hernández-Campos et al., 2025; Rajabalee & Santally, 2021). Despite these theoretical tensions, within the demanding context of high school STEM education, existing literature provides compelling evidence that this construct acts as a primary driver of academic success. For instance, Nia et al. (2023) posited that student satisfaction is not an affective state but a reliable predictor of overall course success, correlating strongly with enhanced academic performance. Complementing this perspective, yet acknowledging that the impact may vary based on task difficulty, Stanislaw (2024) indicated that satisfaction with AI services plays a crucial role in alleviating the cognitive frustration associated with complex topics, sustaining the deep engagement necessary for mastering difficult learning outcomes. Furthermore, Yang et al. (2025) observed that when AI systems meet student needs, the resulting satisfaction fosters a

supportive environment that mitigates anxiety, facilitating the cognitive processes required for achievement. This stands in contrast to systems where technical friction undermines the learning experience (Daud, 2026). Mekheimer (2025) argued that satisfied learners exhibit higher intrinsic motivation and self-efficacy, which drives them to persist through challenges and attain results. Zhang et al. (2025) explained that learning satisfaction optimizes the allocation of cognitive resources toward skill development rather than troubleshooting technical issues, strengthening students' final learning performance. Based on this empirical foundation, the following hypothesis is proposed:

- **H₁₄**: Learning satisfaction has a significant positive influence on learning outcomes.

2.2.9. Research framework

Underpinned by the S-O-R paradigm, the proposed framework for this study (Figure 1) systematically maps how AI tutor characteristics translate into educational results through students' internal psychological processes. In this model, the stimulus is defined by the multidimensional attributes of the AI tutor, encompassing control, efficiency, enjoyment, learning content, personalization, and security, which serve as external environmental triggers within the learning context. These technical and pedagogical features directly impact the organism, representing the learners' internal state, by shaping their cognitive and affective evaluations manifesting as learning motivation and learning satisfaction. Following this internal processing, the framework posits that heightened levels of motivation and satisfaction function as the necessary precursors to the response, which is operationalized here as tangible learning outcomes in STEM subjects. By structuring these relationships sequentially, the model illustrates a clear pathway where the design quality of the AI system fosters positive internal student states that subsequently drive superior academic performance.

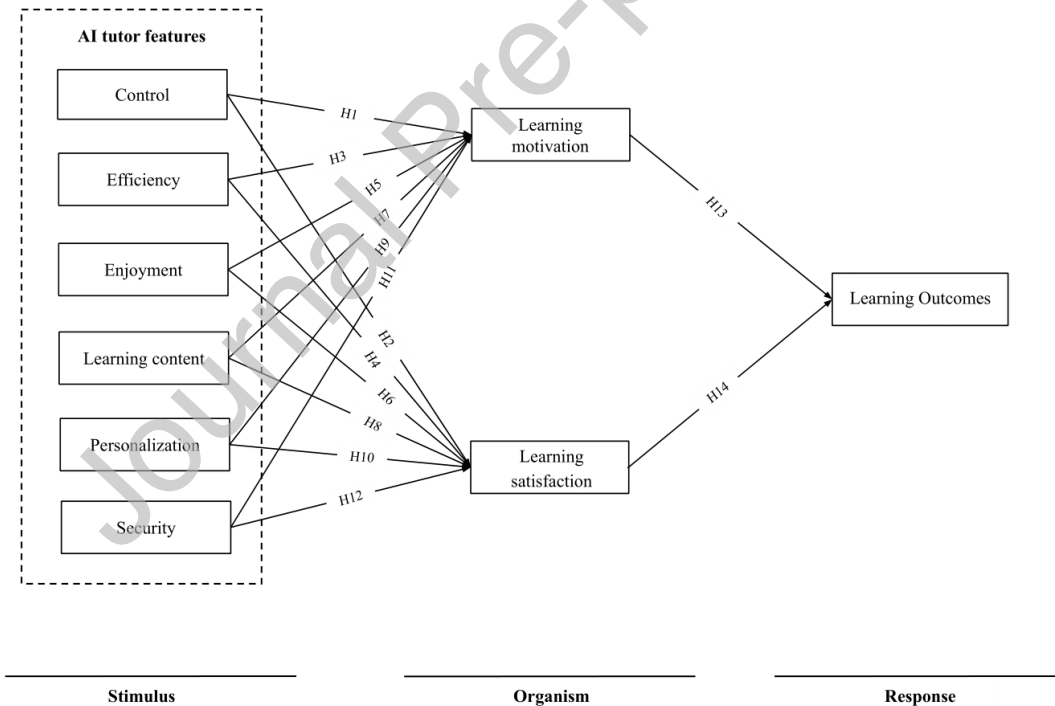


Figure 1. Research framework

3. Methodology

3.1. Questionnaire design

This study adopted a quantitative research design and collected primary data through a self-administered online questionnaire. The survey was developed using Google Forms and was directed at high school students with prior experience using AI tutors for STEM learning. To ensure clarity and logical flow, the questionnaire was organized into four structured stages.

To ensure measurement validity and eliminate potential ambiguity regarding the technological context, the survey commenced with a mandatory introductory section. In this stage, all participants were provided with a comprehensive, standardized definition of “AI tutors” alongside concrete examples of such systems (e.g., adaptive learning platforms and ITS). This section also included detailed explanations of the research variables to distinguish sophisticated AI-driven systems from generic digital resources or basic generative tools. By establishing this shared conceptual framework at the outset, the study minimized the risk of heterogeneous interpretations and ensured that respondents possessed a uniform understanding of the specific tools under investigation.

Following this foundational clarification, the first section functioned as a screening mechanism to identify eligible respondents. Participants were asked preliminary questions such as “Have you ever used an AI tutor for your STEM learning?” and “How often do you use these AI tutors?” Only individuals who indicated prior academic use of AI tutors after reviewing the provided definitions were allowed to proceed, ensuring the relevance, validity, and accuracy of the dataset. The second section collected demographic characteristics and contextual background information. This included variables such as gender, school type, frequency of AI tutor use, and the primary purposes. These data supported the description of the sample profile and provided meaningful contextual insights for subsequent analysis.

The final section contained 36 items measuring the nine latent constructs specified in the theoretical framework. To ensure content validity and enhance construct reliability, all items were adapted from previously validated instruments in the literature. These items were carefully modified to fit the specific context of AI tutors in high school STEM education. The integration of the mandatory definition stage prior to these measurements ensures that the collected responses accurately reflect the intended constructs, thereby maintaining high interpretability and reducing data noise.

The six dimensions of the AI tutor were measured as follows: control items were adapted from Xia et al. (2023); efficiency and security items were drawn from Razak & Khan (2024); enjoyment was measured using scales from Kim et al. (2025) and Rahman et al. (2023); learning content items were based on Kumar et al. (2021) and Sumi & Kabir (2021); and personalization items were adapted from Wang S. et al. (2021). The measures for the remaining constructs were also drawn from prior research, including learning motivation, learning satisfaction, and learning outcomes (Almulla, 2024; Bai & Wang, 2025; Ji et al., 2024; Kim et al., 2014; Kim et al., 2018; Sobaih et al., 2022). All items in this section were evaluated on a 5-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree).

Prior to the main data collection, a pilot test was conducted with 50 high school students to evaluate the clarity, readability, and comprehension of the questionnaire items (Al-Sa'di et al., 2023). The internal consistency reliability of each construct was evaluated using Cronbach's alpha. All constructs demonstrated strong reliability, with Cronbach's Alpha values exceeding the recommended threshold of 0.70 (Khanal & Chhetri, 2024). Feedback from this pilot group led to minor revisions to improve the phrasing of a few items. This pre-testing process ensured the appropriateness and robustness of the measurement instrument before launching the full-scale survey.

3.2. Data collection

This study employed a purposive sampling strategy, a widely accepted non-probability sampling method for research requiring respondents with specific attributes or experiences. This approach was appropriate given the study's objective of obtaining data exclusively from high school students who had prior experience using AI tutors for STEM learning (Stratton, 2023). The selection criteria were defined to ensure that all participants could provide relevant feedback based on their direct interactions with such educational tools (Zamiri & Esmaeili, 2024).

Data were collected entirely through online channels to efficiently reach a digitally connected adolescent population and to access a wide geographical range of schools and communities (Iglesias-Rodriguez et al., 2021). The survey was administered using Google Forms, and the link was disseminated through multiple distribution pathways, including online educational forums and social media platforms such as Facebook, Zalo, and Instagram. This multi-channel approach maximized the likelihood of reaching the intended population while maintaining cost-effectiveness and timeliness.

To ensure the instrument's validity and comprehension, the questionnaire underwent a rigorous development process. It was initially formulated in English based on established literature and subsequently translated into Vietnamese. The research adopted the back-translation methodology recommended by Nguyen (2025) to guarantee linguistic and conceptual equivalence. This involved a forward translation by a bilingual expert, followed by an independent reverse translation by another expert to check for and resolve any semantic discrepancies.

Ethical considerations were strictly adhered to throughout the data collection process, beginning with a clear statement of the research purpose at the start of the survey to inform participants that the information collected would serve solely

for academic research objectives. Informed consent was obtained from all respondents before they proceeded, confirming their voluntary participation and understanding of the study's aims. Data collection was conducted over a period from October 15 to December 1, 2025.

The required sample size was determined based on established guidelines for quantitative research, which recommend a minimum ratio of 10 participants per observed variable to ensure model stability and reliable parameter estimation (Hair & Alamer, 2022). With 36 measurement items in the model, a minimum of 360 valid responses was required. The online survey initially yielded 502 responses. After comprehensive data screening, 478 responses were retained for the final analysis. To ensure the validity of the fully online collection method, responses were excluded if they failed to meet the screening criteria: (1) respondents who were not current high school students, as verified through self-reported demographic questions and consistency checks; (2) individuals who reported no prior use of AI tutors for STEM learning; and (3) incomplete or low-quality responses, including those displaying straight-lining or other indicators of inattentive responding. This rigorous screening process ensured the reliability and integrity of the final dataset.

3.3. Data analysis

To empirically evaluate the hypothesized relationships within the proposed theoretical framework, the study employed PLS-SEM. This analytical technique was selected due to its strengths in handling complex models with multiple latent constructs, its suitability for prediction-oriented and exploratory research designs, and its robustness when working with data that may deviate from multivariate normality (Hair & Alamer, 2022). All statistical analyses were conducted using SmartPLS 3. The significance of model estimates was assessed through a non-parametric bootstrapping procedure with 5,000 resamples and two-tailed tests, allowing for the computation of standard errors, t-statistics, and p-values for all structural paths.

The data analysis procedure followed the established two-step approach recommended in the PLS-SEM literature. The first stage focused on evaluating the measurement model to establish its reliability and validity. Internal consistency was examined using both Cronbach's Alpha (CA) and Composite Reliability (CR). Convergent validity was assessed by inspecting the Average Variance Extracted (AVE), ensuring that each construct explained a sufficient amount of variance from its indicators. Discriminant validity was then confirmed using two distinct methods: the Fornell-Larcker criterion and the Heterotrait-Monotrait (HTMT) ratio. To mitigate the potential influence of common method bias (CMB), this study utilized a multi-faceted validation approach. First, the Variance Inflation Factor (VIF) was calculated for all constructs to monitor potential multicollinearity and provide a statistical assessment of common method bias. VIF values falling below the conservative threshold confirmed that neither multicollinearity nor CMB presented a significant concern for the model's integrity. Following this, the marker variable technique was employed by incorporating "attitude towards aesthetics" into the measurement model. This variable was selected due to its conceptual independence from the study's focal constructs, including control, efficiency, enjoyment, learning content, personalization, security, learning motivation, satisfaction, and outcomes, and was measured using an identical five-point Likert scale. By embedding this marker within the PLS-SEM framework, the correlations between it and the substantive variables were analyzed to identify any remaining evidence of systematic method variance.

In the second stage, the structural model was evaluated to test the proposed hypotheses. This involved assessing the path coefficients (β), t-statistics, and associated p-values generated by the bootstrapping procedure. The model's explanatory strength was determined by examining the R-squared (R^2) values for the endogenous constructs (learning motivation, learning satisfaction, and learning outcomes). The R^2 statistic indicates the percentage of variance in each dependent variable that is explained by its antecedent constructs. Finally, the model's predictive relevance was evaluated using the Stone-Geisser Q^2 statistic, which was obtained through a blindfolding procedure.

4. Results

4.1. Participants' demographic information

A final sample of 478 valid responses from high school students was procured for the final analysis following data screening. Table 1 presents the demographic and behavioral characteristics of the sample. The gender distribution was relatively balanced, with 52.7% male ($n = 252$) and 47.3% female ($n = 226$) respondents. Regarding the educational context, the majority of participants (66.1%, $n = 292$) were enrolled in public high schools, while 33.9% ($n = 186$) attended private institutions. The analysis of AI tutor usage patterns revealed a high level of adoption among respondents. Specifically, 39.7% reported using AI tutors on a daily basis, and 35.4% used them frequently. Only a minority indicated moderate (18.0%), occasional (5.4%), or rare (1.5%) engagement. The duration of each AI tutor interaction also varied across students. The largest proportion (33.7%) reported using AI tutors for 1–2 hours per session, followed by 31.8% who typically engaged for 30–60 minutes. Smaller groups reported session lengths exceeding 2 hours (19.7%) or shorter

than 30 minutes (14.9%). Participants utilized AI tutors for multiple academic purposes, reflecting the diverse functionality of these digital tools. The most common use involved practicing foreign languages or soft skills (29.3%), followed by completing assignments or essays (24.5%) and seeking explanations of complex concepts (19.5%). Additional uses included supporting study planning or resource searching (14.2%) and conducting quick question-and-answer or review sessions (12.6%). Overall, this descriptive profile illustrates that the sampled students represent active and experienced users of AI tutoring systems. Their diverse engagement patterns offer a robust empirical foundation for examining how perceptions of AI tutor features relate to learning motivation, satisfaction, and STEM learning outcomes.

Table 1. Respondents' profile

Demographic Information (N = 478)		Frequency	Percentage
Gender	Male	252	52.7%
	Female	226	47.3%
High schools	Public high school	292	66.1%
	Private high school	186	33.9%
Usage Frequency	Rarely	7	1.5%
	Occasionally	26	5.4%
	Moderately	86	18.0%
	Frequently	169	35.4%
	Daily	190	39.7%
Usage Time	Less than 30 minutes	71	14.9%
	From 30 to 60 minutes	152	31.8%
	From 1 to 2 hours	161	33.7%
	More than 2 hours	94	19.7%
Usage purpose	To explain lessons/concepts	93	19.5%
	To complete assignments/essays	117	24.5%
	To practice foreign languages/soft skills	140	29.3%
	To plan study schedules search for materials	68	14.2%
	For quick Q&A/revision	60	12.6%

4.2. Measurement model assessment

4.2.1. Internal consistency reliability

The assessment of the measurement model began with an evaluation of indicator reliability. For reflective constructs, indicator loadings above 0.70 are considered satisfactory, as they demonstrate that items sufficiently represent their underlying latent variables (Hair et al., 2019). As reported in Table 2, all indicators met this criterion, with loadings exceeding the recommended threshold. The lowest loading was 0.719 (ENJ4), confirming that each item exhibits strong reliability and meaningfully contributes to its respective construct.

Following the confirmation of indicator-level reliability, the analysis proceeded to the construct level by evaluating internal consistency using Cronbach's alpha (CA) and Composite Reliability (CR). Both of these metrics assess the degree to which a set of items cohesively measures a shared underlying construct. The accepted standard for both CA and CR is a value of 0.70 or greater (Hair et al., 2019). The results demonstrate high reliability across all constructs: CA values ranged from 0.740 (ENJ) to 0.835 (SEC), while CR values ranged from 0.837 (ENJ) to 0.890 (SEC). These reliability coefficients not only meet but clearly exceed recommended cutoffs, indicating that the constructs exhibit strong internal consistency.

Table 2. Construct reliability and validity

Constructs' Items	Loadings	CA	CR	AVE
Control (CON)		0.798	0.869	0.623
CON1. "I feel like I have a lot of input in deciding how I use the AI Tutor in my learning."	0.824			
CON2. "I feel a sense of freedom when using the AI tutor."	0.785			
CON3. "I have many opportunities with the AI tutor to decide for myself how to learn."	0.802			
CON4. "I have a say regarding what input I want to learn with AI tutors."	0.745			
Efficiency (EFF)		0.754	0.844	0.575
EFF1. "The AI tutor works correctly on the first attempt."	0.784			
EFF2. "I can complete tasks with the AI tutor in a short time."	0.723			
EFF3. "The AI tutor's interface presents information clearly."	0.721			
EFF4. "The AI tutor adequately meets my learning requirements."	0.803			
Enjoyment (ENJ)		0.740	0.837	0.561
ENJ1. "I am interested in using the AI tutor for learning"	0.753			
ENJ2. "I feel amazing when I use the AI tutor."	0.767			
ENJ3. "The actual process of learning with the AI tutor is pleasant."	0.756			
ENJ4. "I have fun while using the AI tutor for learning."	0.719			
Learning Content (LCT)		0.784	0.861	0.607
LCT1. "The learning content provided by the AI tutor is well-structured."	0.757			
LCT2. "The learning materials from the AI tutor are up-to-date."	0.821			
LCT3. "The AI tutor provides sufficient learning materials for my study."	0.750			
LCT4. "The AI tutor offers content that is necessary and useful for my learning."	0.786			
Personalization (PER)		0.832	0.888	0.664
PER1: "The personalized learning content recommended by AI tutors is what I am interested in."	0.823			
PER2: "The personalized learning content recommended by AI tutors is what I like to learn."	0.818			
PER3: "The personalized learning content recommended by AI tutors is based on my needs."	0.803			

PER4: “The personalized learning content recommended by AI tutors is tailored to my situation.” 0.816

Security (SEC)	0.835	0.890	0.668
SEC1. “There is no risk of personal information loss when disclosing it to the AI tutor.”	0.791		
SEC2. “I feel secure in providing sensitive information to the AI tutor.”	0.816		
SEC3. “I believe that the AI tutor protects my personal data.”	0.823		
SEC4. “I trust that my personal information will not be misused by the AI tutor.”	0.839		
Learning Motivation (LMO)	0.815	0.878	0.643
LMO1. “I feel more motivated to engage in learning activities with the support of the AI tutor.”	0.797		
LMO2. “Interacting with the AI tutor stimulates my motivation for focused and effective learning.”	0.812		
LMO3. “Even when studying is hard, the AI tutor helps me figure out a way to keep going.”	0.818		
LMO4. “If studying gets too boring, the AI tutor helps me find a way to make it fun.”	0.782		
Learning Satisfaction (LSA)	0.824	0.883	0.654
LSA1. “The learning opportunities provided by the AI tutor contribute to my overall satisfaction.”	0.796		
LSA2. “I am highly satisfied with the learning outcomes facilitated by the AI tutor.”	0.814		
LSA3. “I have no complaints about using the AI tutor for learning.”	0.799		
LSA4. “I am satisfied with my overall experience of learning with the AI tutor.”	0.825		
Learning Outcomes (LOC)	0.793	0.865	0.617
LOC1. “Using the AI tutor has improved my learning.”	0.767		
LOC2. “The AI tutor supports me in learning more.”	0.774		
LOC3. “The AI tutor supports me to continue learning.”	0.799		
LOC4. “Using the AI tutor has allowed me to better understand the concepts studied.”	0.800		

Notes. CA = Cronbach’s Alpha; CR = Composite Reliability; AVE = Average Variance Extracted

4.2.2. Convergent validity

Convergent validity was assessed to determine whether indicators associated with each latent construct converge to measure the same underlying concept. The Average Variance Extracted (AVE) was used for this purpose, representing the proportion of variance captured by a construct relative to measurement error. According to established methodological standards, an AVE value of 0.50 or higher is the recommended threshold for adequacy (Hair & Alamer, 2022). An AVE at this level signifies that the latent variable explains, on average, at least half of the variance in its corresponding observed indicators. The AVE values calculated for this study ranged from 0.561 (ENJ) to 0.668 (SEC),

all of which exceeded the recommended threshold (Table 2). These results confirm that the constructs demonstrate strong convergent validity and that their indicators share substantial common variance.

4.2.3. Discriminant validity

Discriminant validity was evaluated to ensure that each construct in the research model is empirically distinct and measures a unique concept. This step is crucial to ensure that the findings are not confounded by overlapping variables (Cheung et al., 2024). This study employed two established methods to evaluate discriminant validity: the Fornell-Larcker criterion and the HTMT ratio of correlations.

First, the Fornell–Larcker criterion was applied by comparing the square root of each construct’s AVE with its correlations with other constructs. As shown in Table 3, the square roots of the AVE (e.g., CON = 0.789; EFF = 0.759; ENJ = 0.749) consistently exceed the corresponding inter-construct correlation coefficients in their corresponding rows and columns. This indicates that constructs share more variance with their own indicators than with other constructs, thereby satisfying the criterion (Fornell & Larcker, 1981).

Second, the HTMT ratio was calculated to provide a more rigorous assessment of discriminant validity. Following the guidelines established by Roemer et al. (2021), this study adopted the conservative threshold of 0.90. The results detailed in Table 4 show that all HTMT values are below this 0.90 threshold. The highest observed ratio was 0.857 (between PER and LOC), which remains within the acceptable range. This comprehensive assessment confirms that multicollinearity is not a significant concern and that all constructs are sufficiently distinct from one another. Overall, the combined evidence from both the Fornell–Larcker test and the HTMT analysis provides strong support for the discriminant validity of the measurement model, confirming the conceptual distinctiveness of all constructs included in this study.

Table 3. Fornell-Larcker criterion

	CON	EFF	ENJ	LCT	LMO	LOC	LSA	PER	SEC
CON	0.789								
EFF	0.414	0.759							
ENJ	0.505	0.612	0.749						
LCT	0.603	0.460	0.489	0.779					
LMO	0.537	0.521	0.557	0.590	0.802				
LOC	0.582	0.478	0.537	0.607	0.685	0.785			
LSA	0.579	0.493	0.554	0.604	0.695	0.660	0.809		
PER	0.561	0.421	0.500	0.645	0.621	0.697	0.629	0.815	
SEC	0.644	0.417	0.508	0.627	0.621	0.624	0.603	0.577	0.818

Notes. CON = Control; EFF = Efficiency; ENJ = Enjoyment; LCT = Learning Content; PER = Personalization; SEC = Security; LMO = Learning Motivation; LSA = Learning Satisfaction; LOC = Learning Outcomes

Table 4. Heterotrait-Monotrait (HTMT) ratio

	CON	EFF	ENJ	LCT	LMO	LOC	LSA	PER	SEC
CON									
EFF	0.529								
ENJ	0.655	0.813							
LCT	0.761	0.598	0.637						
LMO	0.666	0.661	0.713	0.731					
LOC	0.732	0.615	0.699	0.767	0.850				

LSA	0.712	0.623	0.707	0.747	0.848	0.815		
PER	0.690	0.531	0.635	0.801	0.751	0.857	0.760	
SEC	0.789	0.524	0.643	0.769	0.752	0.765	0.727	0.691

Notes. CON = Control; EFF = Efficiency; ENJ = Enjoyment; LCT = Learning Content; PER = Personalization; SEC = Security; LMO = Learning Motivation; LSA = Learning Satisfaction; LOC = Learning Outcomes

4.3. Structural model assessment

4.3.1. Assessment of common method bias (CMB)

To safeguard the structural model's integrity against potential CMB, a rigorous dual-diagnostic approach was implemented, comprising a full collinearity assessment and the marker variable technique (Hulland et al., 2018). The former examines the inner VIF across all latent constructs to detect systemic bias, while the latter involves the integration of a theoretically irrelevant variable to quantify potential method variance. Following established benchmarks, CMB is considered negligible if inner VIF values remain below the 3.3 threshold and the marker variable demonstrates no substantial correlation with the primary constructs (Kock, 2015).

As presented in Table 6, the results of the full collinearity test indicate that inner VIF values for all predictors ranged from 1.701 to 2.242. Since these values are well within the conservative limit of 3.3, it is confirmed that the dataset is free from problematic multicollinearity and that common method variance is unlikely to distort the findings. These results were further validated through the marker variable technique, utilizing "attitude towards aesthetics" as a theoretically distinct construct. This variable exhibited only negligible and statistically non-significant correlations with all substantive variables (all $r < 0.10$, $p > 0.05$). Furthermore, the structural relationships between AI tutor features, motivation, and learning outcomes remained stable upon the inclusion of this marker. The assessment was concluded by comparing the R^2 values of the key endogenous variables before and after the introduction of the marker variable. This analysis revealed minimal variance: R^2 for learning outcomes increased by only 0.1% (from 0.535 to 0.536), learning satisfaction rose by 0.2% (from 0.552 to 0.554), and learning motivation remained constant at 0.554. Given that these fluctuations are substantially lower than the 10% maximum suggested by Gaskin et al. (2025), it is concluded that common method bias does not pose a threat to the validity of the hypothesized psychological pathways in this STEM-focused study.

4.3.2. Results of testing hypotheses

The empirical results from the hypothesis testing, detailed in Table 5 and visually represented in the structural model in Figure 2, provide comprehensive evidence regarding the influence of AI tutor features on students' motivation, satisfaction, and learning outcomes in STEM education. The analysis yields significant support for the majority of the hypothesized relationships. This support is determined by evaluating the path coefficients (beta), their corresponding t statistic, and significance levels (p-values). Generally, a hypothesis is accepted as statistically significant if its p-value falls below a predetermined threshold of 0.05, or correspondingly, if its t statistic exceeds a critical threshold of 1.96 for a 95% confidence level, which indicates that the observed effect is unlikely to have occurred by chance (Hair et al., 2019).

The investigation into the antecedents of student learning experiences reveals robust support for nearly all proposed AI tutor dimensions. A notable exception was found in the path from control to learning motivation (H1: $\beta = 0.034$, $t = 0.659$, $p = 0.510$), which was not statistically significant. This suggests that while students may use control features, it does not directly translate into increased motivation. However, control did demonstrate a significant positive influence on learning satisfaction (H2: $\beta = 0.127$, $t = 2.610$, $p = 0.009$), indicating that the ability to manage the learning pace or interface contributes to a more satisfactory experience.

Conversely, efficiency (H3: $\beta = 0.166$, $t = 3.573$, $p = 0.000$; H4: $\beta = 0.114$, $t = 2.476$, $p = 0.013$), enjoyment (H5: $\beta = 0.132$, $t = 2.690$, $p = 0.007$; H6: $\beta = 0.139$, $t = 2.710$, $p = 0.007$), and learning content (H7: $\beta = 0.111$, $t = 2.002$, $p = 0.045$; H8: $\beta = 0.135$, $t = 2.499$, $p = 0.012$) all emerged as significant positive predictors for both learning motivation and learning satisfaction. These findings confirm that a seamless, engaging, and content-rich AI tutor environment is foundational to fostering positive student attitudes.

Among all AI tutor features, personalization and security exhibited the most substantial impacts. Personalization was a particularly strong predictor for both learning motivation (H9: $\beta = 0.252$, $t = 4.489$, $p = 0.000$) and learning satisfaction (H10: $\beta = 0.255$, $t = 4.077$, $p = 0.000$). This highlights the critical importance of adaptive learning paths and tailored

feedback. Similarly, security, which relates to data privacy and a safe learning environment, was a powerful driver for learning motivation (H11: $\beta = 0.248$, $t = 4.528$, $p = 0.000$) and learning satisfaction (H12: $\beta = 0.172$, $t = 3.339$, $p = 0.001$).

Regarding the subsequent impact on learning outcomes, the analysis provides strong empirical validation for the final two hypotheses. Both learning motivation (H13: $\beta = 0.437$, $t = 7.790$, $p = 0.000$) and learning satisfaction (H14: $\beta = 0.357$, $t = 6.439$, $p = 0.000$) were found to be highly significant positive predictors of learning outcomes. The coefficient values indicate that these two constructs are the most powerful drivers of academic success within the model.

In summary, the findings validate 13 of the 14 proposed hypotheses. The results particularly emphasize the pronounced roles of personalization and security in shaping students' internal motivational and satisfaction states. Furthermore, the analysis confirms that learning motivation and learning satisfaction are pivotal drivers that ultimately contribute to enhanced learning outcomes in the context of AI-powered STEM tutoring.

Table 5. Hypothesis testing result

H	Paths	Coefficient (β)	Sample Mean	Standard Deviation	T Statistic	P-Values	VIF	Results
H1	CON $\rightarrow$ LMO	0.034	0.036	0.051	0.659	0.510	2.053	Rejected
H2	CON $\rightarrow$ LSA	0.127	0.126	0.049	2.610	0.009	2.053	Accepted
H3	EFF $\rightarrow$ LMO	0.166	0.167	0.047	3.573	0.000	1.701	Accepted
H4	EFF $\rightarrow$ LSA	0.114	0.114	0.046	2.476	0.013	1.701	Accepted
H5	ENJ $\rightarrow$ LMO	0.132	0.132	0.049	2.690	0.007	1.949	Accepted
H6	ENJ $\rightarrow$ LSA	0.139	0.141	0.051	2.710	0.007	1.949	Accepted
H7	LCT $\rightarrow$ LMO	0.111	0.112	0.055	2.002	0.045	2.242	Accepted
H8	LCT $\rightarrow$ LSA	0.135	0.136	0.054	2.499	0.012	2.242	Accepted
H9	PER $\rightarrow$ LMO	0.252	0.252	0.056	4.489	0.000	2.001	Accepted
H10	PER $\rightarrow$ LSA	0.255	0.252	0.062	4.077	0.000	2.001	Accepted
H11	SEC $\rightarrow$ LMO	0.248	0.244	0.055	4.528	0.000	2.154	Accepted
H12	SEC $\rightarrow$ LSA	0.172	0.172	0.052	3.339	0.001	2.154	Accepted
H13	LMO $\rightarrow$ LOC	0.437	0.439	0.056	7.790	0.000	1.933	Accepted
H14	LSA $\rightarrow$ LOC	0.357	0.355	0.055	6.439	0.000	1.933	Accepted

Note. CON = Control; EFF = Efficiency; ENJ = Enjoyment; LCT = Learning Content; PER = Personalization; SEC = Security; LMO = Learning Motivation; LSA = Learning Satisfaction; LOC = Learning Outcomes

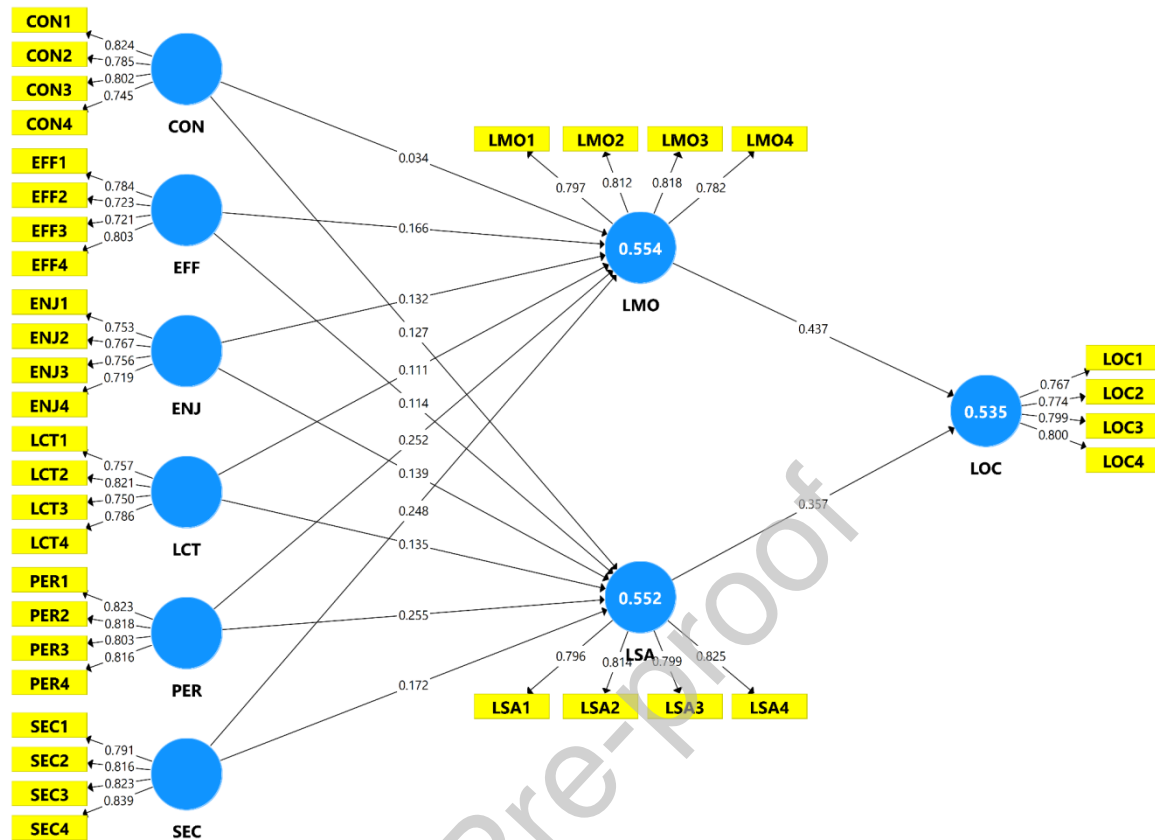


Figure 2. Results of PLS-SEM analysis

4.3.3. Evaluation of explanation and prediction power of research model

To further assess the robustness of the structural model, both its explanatory capacity and predictive relevance were examined using the Coefficient of Determination (R^2) and the Stone–Geisser Q^2 statistic. These metrics are widely recognized as essential criteria for evaluating the quality of prediction-oriented PLS-SEM models.

The R^2 values indicate the proportion of variance in each endogenous construct that is accounted for by its respective predictors. According to established benchmarks, R^2 values of 0.75, 0.50, and 0.25 reflect substantial, moderate, and weak explanatory power, respectively (Sarstedt et al., 2021). In this study, the AI tutor dimensions collectively explained 55.4% of the variance in learning motivation ($R^2 = 0.554$) and 55.2% of the variance in learning satisfaction ($R^2 = 0.552$). Moreover, learning motivation and learning satisfaction jointly accounted for 53.5% of the variance in learning outcomes ($R^2 = 0.535$). With all R^2 values exceeding the 0.50 threshold, the structural model demonstrates moderate to substantial explanatory power, indicating that the identified antecedents meaningfully account for variations in students' motivational, affective, and academic responses.

Predictive relevance was then evaluated using the Q^2 statistic generated through the blindfolding procedure. A Q^2 value greater than zero indicates that the model exhibits predictive relevance for a given construct (Sharma et al., 2021). The results provide compelling evidence of strong predictive performance: Q^2 values were 0.536 for learning motivation, 0.533 for learning satisfaction, and 0.549 for learning outcomes. These values substantially exceed the minimum threshold, confirming that the model provides accurate out-of-sample predictions for all endogenous variables.

Taken together, the R^2 and Q^2 results affirm that the proposed research model is not only theoretically sound but also empirically robust. The model demonstrates meaningful explanatory capacity and high predictive relevance, underscoring its suitability for advancing understanding of how AI tutor features shapes student motivation, satisfaction, and learning outcomes in STEM education.

Table 6. Model performance metrics

Constructs	R^2	R^2 Adjusted	Q^2
------------	-------	----------------	-------

LMO	0.554	0.548	0.536
LSA	0.552	0.546	0.533
LOC	0.535	0.533	0.549

Notes. LMO = Learning Motivation; LSA = Learning Satisfaction; LOC = Learning Outcomes

5. Discussion

5.1. Impacts of control on learning motivation and learning satisfaction

The analysis of AI tutor dimensions provides important and nuanced insights into how high school students engage with AI-mediated STEM learning. One of the most striking findings concerns the dimension of control. Contrary to theoretical expectations, the results indicated that perceived control did not significantly influence learning motivation, leading to the rejection of Hypothesis H1. This suggests that allowing students to adjust the pace, sequence, or focus of their AI-supported lessons does not inherently enhance their intrinsic drive to engage with STEM subjects. This result presents a notable divergence from the theoretical groundwork, particularly the core tenets of Self-Determination Theory (SDT), which posits that autonomy is a fundamental psychological need essential for fostering intrinsic motivation (Ruiz & Yabut, 2024). It also contrasts with prior empirical studies showing that higher levels of perceived control enhance intrinsic interest, persistence, and engagement in digital and STEM-focused learning environments (Bognár et al., 2024; Guo et al., 2025). Furthermore, this finding is inconsistent with the work of Xu et al. (2024), which highlights the motivational benefits of self-regulated learning features. The reasons for this discrepancy likely lie in the cultural and contextual moderators inherent to Vietnam's exam-oriented education system and its Confucian-influenced socio-cultural framework. From the perspective of Self-Determination Theory, while the AI tutor provides autonomy-supportive features through user control, these features may not be powerful enough to override the dominant extrinsic regulatory structures that define the lives of Vietnamese adolescents. In Vietnam, the "exam preparation" culture creates a unique psychological environment where academic success is viewed not only as an individual achievement but also as a fulfillment of collective family obligations. In this high-stakes testing culture, motivation is predominantly anchored in performance-approach goals, driven by the necessity of passing university entrance examinations and meeting high parental expectations, rather than the internal desire for autonomous exploration. Consequently, while students may utilize control features to navigate content more efficiently, these tools do not necessarily catalyze a shift from extrinsic to intrinsic motivation. This suggests that in highly hierarchical and structured educational environments, the mere presence of autonomy-enabled technology is insufficient to alter deep-seated, performance-based motivational structures that prioritize "what is required" over "what is personally interesting." Therefore, the applicability of Western-centric autonomy-motivation links may be limited in developing Asian contexts where institutional pressures and cultural norms regarding academic duty outweigh the perceived value of instructional agency.

In contrast, the study found strong support for Hypothesis H2, confirming that control is a significant positive predictor of learning satisfaction. This result indicates that when students feel empowered to direct their educational journey, such as by adjusting instructional difficulty or revisiting complex topics, they report a substantially higher level of overall satisfaction with the AI tutor service. This outcome is highly consistent with the existing literature demonstrating that learner control enhances satisfaction in technology-mediated learning environments (Liang et al., 2024). Studies in AI-based and multimedia learning suggest that control reduces cognitive load, improves navigation efficiency, and reduces frustration, thereby contributing to a smoother and more satisfying experience (Gkintoni et al., 2025). Furthermore, it supports the work of Tan et al. (2025), who linked the ability to personalize the learning path to a more positive and seamless user experience. The divergence between the rejection of H1 and the acceptance of H2 provides a critical theoretical insight: autonomy-supportive features in AI tutors can enhance the "quality of the experience" (satisfaction) without necessarily increasing the "will to learn" (motivation). This distinction is particularly relevant to the technology adoption patterns observed among Vietnamese high schoolers, who tend to adopt a pragmatic approach to digital tools, viewing them primarily as instruments for efficiency rather than exploration. In the Vietnamese STEM context, where students frequently face exceptionally dense curricula and rapid instructional pacing, the ability to control the interaction functions as a vital coping mechanism. By allowing students to revisit difficult material or adjust the speed of content delivery, the AI tutor provides psychological reassurance and mitigates the acute stress associated with high-stakes assessments. Thus, control facilitates a more comfortable learning process within a rigid institutional framework. For high school students in an exam-heavy culture, managing the interaction is fundamental to their positive evaluation of the tool (satisfaction), even if the overarching "extrinsic" pressure of the education system prevents that control from translating into "intrinsic" motivation. These findings underscore the necessity of recognizing cultural and institutional boundaries; design principles for AI tutors that emphasize control may function as satisfaction-boosters in high-pressure educational systems, but they should not be mistaken for universal catalysts of intrinsic motivation in environments where academic goals are culturally and socially prescribed.

5.2. Impacts of efficiency on learning motivation and learning satisfaction

The results of this study demonstrate that efficiency exerts a strong and positive effect on students' learning motivation in STEM, thereby supporting Hypothesis H3. This indicates that when an AI tutor optimizes the learning journey by reducing operational effort, organizing learning pathways, and facilitating rapid access to instructional resources, students experience a greater internal drive to engage with STEM tasks. This finding is consistent with the work of Chisunum and Nwadiokwu (2024), who found that effective time utilization enhances learners' academic engagement. The finding also reinforces the perspectives of Gavrilas and Kotsis (2025) and Gkintoni et al. (2025), who argued that efficient learning systems mitigate extraneous cognitive load. By doing so, the AI tutor allows students to dedicate their mental resources to mastering complex STEM concepts rather than to navigating the system itself, a process that sustains their motivation. This study's finding also resonates with the work of Tavdgiridze et al. (2024) on the importance of immediate and targeted feedback, a key component of efficiency that accelerates comprehension and fosters a positive feedback loop of success and continued effort, as also reflected in Bejtlic's (2024) conceptualization of motivational reinforcement. While these results generally accord with prior research, they are deeply rooted in the Vietnamese high school STEM context and may not be universally applicable without considering specific boundary conditions. Vietnamese students operate within a socio-cultural framework characterized by high educational pressure and a "result-oriented" learning culture, often driven by the high stakes of national examinations. In this environment, efficiency is not merely a technical convenience but a psychological necessity. Vietnamese students often face heavy curriculum loads and limited class time for inquiry-based learning; therefore, an AI tutor that reduces time-consuming trial-and-error processes holds disproportionately high motivational value. This suggests that the relationship between efficiency and motivation is moderated by the level of academic competition and institutional pressure inherent in the Vietnamese system, where saving time directly correlates with reduced anxiety and increased perceived competence. Consequently, these findings should be interpreted within the context of developmental and emerging economies where educational resources are specialized and standardized testing remains a primary driver of student behavior.

In addition, the results also confirm that efficiency is a vital determinant in enhancing high school students' learning satisfaction with AI tutors in STEM education, thus validating Hypothesis H4. This indicates that when students perceive the AI tutor as seamless, responsive, and intuitive, they report a greater sense of contentment with their educational experience. This outcome is highly consistent with established frameworks, such as the Technology Acceptance Model (TAM), which positions perceived ease of use, a concept closely related to efficiency, as a fundamental antecedent of user satisfaction (Zhou et al., 2022). The finding also echoes the conclusions of Liu and Liu (2025), who noted that systems reducing operational complexity are met with a more positive affective response from users. Moreover, this result aligns with Pandita and Kiran (2023), who highlighted that minimizing user frustration is critical for a positive interaction. Within the Vietnamese high school context, this finding underscores a unique nuance regarding technology adoption patterns. While Vietnamese students are increasingly digital natives with high personal smartphone penetration, school technology infrastructure remains inconsistent across different provinces. The findings suggest that students are more satisfied when the AI tutor compensates for these infrastructural gaps by providing a smooth, low-latency, and reliable user experience that functions well even under suboptimal conditions. Culturally, Vietnamese learners tend to exhibit a pragmatic approach to technology adoption, prioritizing utility and reliability over aesthetic novelty. Therefore, any technical barrier, such as slow response times or rigid navigation, is viewed not just as a minor inconvenience but as a failure of the tool to fulfill its primary purpose of academic support. However, it is important to note that these satisfaction levels may differ in Western educational settings where pedagogical norms might prioritize exploratory, open-ended system interactions over the streamlined, efficiency-first design preferred by the Vietnamese cohorts. Thus, while efficiency is a central requirement for satisfaction in this study, its influence may be bounded by the specific institutional and cultural expectations of the Vietnamese secondary education sector.

5.3. Impacts of enjoyment on learning motivation and learning satisfaction

The results offer clear support for Hypothesis H5, demonstrating that enjoyment has a substantial positive effect on high school students' learning motivation when engaging with AI tutors in STEM subjects. This underscores that when students perceive their interactions with the AI tutor as intrinsically rewarding and engaging, their internal drive to learn the subject matter is significantly strengthened. Given that STEM subjects often provoke anxiety or disengagement among adolescents, an enjoyable learning experience appears to serve as an essential counterbalance, reducing negative emotions and reinforcing participation. These findings are strongly aligned with previous empirical studies. For instance, Ortiz-Rojas et al. (2025) emphasized that enjoyable or gamified learning environments enhance students' intrinsic motivation in STEM. It also aligns with the conceptualization of enjoyment as a state of accomplished engagement that fosters resilient learning behaviors and enables students to build cognitive resources (Bustamante-Mora et al., 2025; Mierzwa-Kamińska, 2025). However, the interpretation of these results must be situated within the specific socio-cultural boundaries of the Vietnamese educational system. Vietnam's learning culture is deeply rooted in Confucian values, which prioritize high-stakes examination performance and academic rigor over the affective quality of the

learning process. Consequently, Vietnamese high school students typically face an immense academic workload and intense societal pressure to succeed in STEM disciplines. In this context, the psychological impact of enjoyment may be more pronounced than in Western, student-centered environments. For these students, the AI tutor acts as a critical affective buffer; by transforming demanding, abstract concepts into less intimidating and more appealing tasks, the technology provides a rare source of intrinsic pleasure in an otherwise high-pressure environment. Therefore, while the motivational benefits of enjoyment are theoretically universal, their role as a compensatory mechanism against academic burnout is a distinct characteristic of the Vietnamese high-pressure context. This suggests a cultural boundary: the strength of the enjoyment-motivation link may vary in societies where educational stress is less centralized or where emotional well-being is already a core component of the standard curriculum.

This study also provides support for hypothesis H6, demonstrating that enjoyment is a significant positive predictor of learning satisfaction. This outcome highlights that a student's overall satisfaction with an AI tutor is not merely a functional assessment of its accuracy but is also deeply rooted in their affective experience. A positive emotional response during the learning process appears to be a fundamental component of a student's holistic and subjective appraisal of the service. This outcome resonates with prior research in e-learning, which has consistently shown that a more enjoyable process leads to greater user satisfaction, positioning satisfaction as an evaluation of the entire learning journey (Martin & Bolliger, 2022; Zardari et al., 2021). Furthermore, the result aligns with the proposition that system features cultivating a positive emotional state, such as interactive simulations, serve as a critical lens through which students evaluate the educational tool (Yang et al., 2025). As suggested by Guo et al. (2025), enjoyment functions as a key mechanism that translates perceived system quality into a positive subjective evaluation. The alignment of these studies with the present findings is further explained by specific technology adoption patterns in Vietnam. As a rapidly developing digital economy, Vietnam has seen a surge in "digital-native" adolescents who are highly accustomed to sophisticated, interactive, consumer-grade platforms. Their expectations for educational technology are consequently shaped by their experiences with non-academic digital entertainment. In this specific developmental and institutional setting, students do not view "enjoyment" as a secondary feature but as a primary benchmark for system quality. An AI tutor that fails to provide an engaging interface may be perceived as a burdensome extension of traditional, monotonous schoolwork, leading to rapid disengagement. Thus, the satisfaction of Vietnamese students is contingent upon the technology's ability to bridge the gap between rigorous academic requirements and modern interactive standards. These findings imply that design principles for AI tutors should not be treated as "one-size-fits-all" universalities; instead, they must account for the local digital maturity and the specific psychological expectations of the target student population to ensure long-term adoption and efficacy.

5.4. Impacts of learning content on learning motivation and learning satisfaction

The findings provide robust empirical support for Hypothesis H7, demonstrating that the quality of learning content significantly enhances high school students' learning motivation in STEM subjects. This indicates that when AI tutors deliver materials perceived as relevant, accurate, and intellectually engaging, students exhibit a stronger intrinsic drive to engage with the subject matter. These results align with prior studies emphasizing the centrality of content in driving academic engagement. For instance, Pellas (2025) argued that dynamic and interactive materials, such as virtual simulations, are crucial for sparking curiosity in abstract scientific subjects. These findings also align with the work of Bansal et al. (2023) and Wolf and Ebner (2025), who suggested that varied and well-structured instructional materials are essential for reducing the monotony often associated with traditional study methods and sustaining student persistence. However, the intensity of this motivational impact must be interpreted through the specific lens of the Vietnamese educational landscape. In Vietnam, high school STEM education is characterized by extreme academic competitiveness and a deep-seated cultural emphasis on scholarship as a primary vehicle for social mobility. Consequently, the "motivation" observed in this study likely reflects a synthesis of intrinsic interest and a pragmatic response to high-pressure environmental demands. While the pedagogical clarity of AI-delivered content helps bridge understanding gaps as suggested by the data, its role is amplified by the fact that Vietnamese students often seek supplementary AI tools to master a rigorous, standardized curriculum that leaves little room for error in high-stakes examinations. Therefore, the motivational power of content quality here is not merely an affective preference but a functional necessity for navigating an environment defined by high educational pressure. These findings may be most applicable to similar Confucian-influenced cultures or centralized educational systems where curriculum adherence and academic achievement are paramount and may differ in contexts where inquiry-based or low-stakes learning environments are the norm.

The study also confirms Hypothesis H8, showing that learning content significantly influences learning satisfaction. This indicates that students' overall evaluation of the AI tutor is heavily contingent on the perceived quality, clarity, and pedagogical effectiveness of the instructional materials. This finding is consistent with a broad literature base asserting that the substance of an educational tool is a primary determinant of user satisfaction. Gavrilas and Kotsis (2025), for example, highlighted that clear, logically structured content is essential for reducing cognitive friction, allowing students

to focus on understanding complex principles. This result also reinforces the idea that high-quality content enhances perceived learning effectiveness; as noted by Khairuddin et al. (2024), when students feel they are making tangible progress due to reliable materials, their satisfaction improves. This aligns with the criteria set forth by Ho et al. (2021), who identified material sufficiency, relevance, and accuracy as key determinants. In the Vietnamese context, satisfaction is intrinsically tied to the practical utility of the tool in achieving measurable academic success. Vietnamese technology adoption patterns in education are often “utility-first,” where students and parents prioritize tools that offer clear alignment with national examination standards over those that offer purely aesthetic or social features. The high satisfaction scores related to content quality reflect a cultural norm where the teacher, or in this case, the AI tutor, is primarily valued for their expertise and ability to simplify complex, exam-oriented knowledge. It is critical to acknowledge that these results might be context-specific; the primacy of content over personalization or enjoyment might be less pronounced in Western educational settings that prioritize student autonomy and diverse learning pathways over standardized performance. Thus, while the findings offer a blueprint for AI tutor design, they should be viewed as principles optimized for goal-oriented, high-pressure STEM environments rather than universal axioms for all developmental or institutional frameworks.

5.5. Impacts of personalization on learning motivation and learning satisfaction

The results provide strong empirical support for Hypothesis H9, demonstrating that personalization within the AI tutor significantly enhances learning motivation. Unlike a standardized approach, the system’s ability to dynamically adapt instructional pathways and content pacing does more than simply deliver knowledge; it fundamentally alters the student’s psychological engagement with complex STEM subjects. By calibrating the difficulty level to match individual proficiency, the AI tutor effectively mitigates the dual risks of anxiety from overly difficult tasks and boredom from trivial ones, an empirical validation of the adaptive mechanisms theoretically proposed by Ali and Elkot (2025) and Zhai and Krajcik (2024). This finding implies that the core driver of motivation is the shift from a passive reception of information to an active, tailored learning experience where students feel a sense of competence and control. Within the Vietnamese socio-cultural framework, which is deeply rooted in Confucian-heritage values emphasizing academic achievement and a traditionally teacher-centered pedagogy, this impact is particularly salient. In such environments, the AI tutor serves as a vital pedagogical shift, moving away from the “one-size-fits-all” model often necessitated by large class sizes and a rigid national curriculum. For Vietnamese students, who operate under significant socio-cultural pressure to excel, personalization transforms STEM learning from a potentially overwhelming obligation into an accessible and manageable journey. Furthermore, this positive motivational response reflects specific technology adoption patterns in Vietnam, where digital tools are increasingly viewed by both parents and students as essential “competitive boosters” rather than mere supplements, thus heightening the perceived value of individualized instructional pathways.

The study further confirms Hypothesis H10, demonstrating a significant positive relationship between personalization and learning satisfaction. This implies that students’ overall appraisal of AI tutors is strongly influenced by the system’s capacity to deliver a responsive, user-centric experience. When the AI tutor tailors explanations of intricate scientific concepts or mathematical procedures to a student’s comprehension level, it enhances both the perceived quality and relevance of the interaction. This result corroborates prior evidence emphasizing the importance of individualized feedback in STEM contexts. For example, Villegas-Ch et al. (2025) underscored that targeted guidance addressing specific misconceptions is more effective than generic corrections, while Maier and Klotz (2022) demonstrated that students receiving personalized feedback report significantly higher satisfaction levels. Moreover, responsiveness conveys a sense of recognition and support, which is particularly salient for high-achieving or exam-oriented students who may experience high cognitive demands (Glick et al., 2024; Tan et al., 2025). However, it is crucial to interpret these findings within the specific institutional and developmental boundaries of the Vietnamese high school context. The high level of satisfaction reported may be inherently tied to the intense academic pressure of the secondary education cycle, where students prioritize efficiency and “score-maximization” in preparation for high-stakes national examinations. Consequently, while these results offer valuable insights, they may not be universally applicable to educational systems with lower academic pressure or those that prioritize exploratory, open-ended learning over standardized outcomes. In contexts where cultural norms place less emphasis on hierarchical achievement, the psychological pathway from personalization to satisfaction might be moderated by different developmental factors, such as the need for social collaboration rather than individual mastery. By situating these findings within the Vietnamese context, this study acknowledges that while personalization is a powerful design principle, its effectiveness is deeply intertwined with the specific institutional pressures and cultural expectations of the learners.

5.6. Impacts of security on learning motivation and learning satisfaction

The confirmation of Hypothesis H11 demonstrates that security is not merely a technical safeguard but a psychological prerequisite for learning motivation among high school students in STEM. Rather than simply aligning with general

technology acceptance models, the data suggests that for adolescents engaging with complex STEM subjects, the perception of an AI tutor as a safe zone directly influences their willingness to exert cognitive effort. While Al-Zahrani and Alasmari (2024) identify security as an antecedent for motivation, this study's findings clarify the mechanism: a secure platform minimizes privacy anxiety, thereby liberating cognitive resources that would otherwise be spent on defensive monitoring of personal data. As indicated by Popescu et al. (2024), this reduction in perceived risk is crucial; it transforms the AI tutor from a potential surveillance tool into a trusted partner, encouraging students to experiment with novel problem-solving strategies without fear of exposure or judgment. Conversely, as Habbal et al. (2024) warn, the absence of this trust fosters inhibition, effectively neutralizing the pedagogical benefits of the AI interaction. However, the magnitude of this psychological pathway must be interpreted within the specific Vietnamese socio-cultural context, which is characterized by intense academic pressure and a high-stakes examination culture. In Vietnam, where STEM subjects are often the focal point of fierce peer competition and parental expectations, the fear of “losing face” through public academic struggle is a significant inhibitor. Consequently, the AI tutor's security features do not merely fulfill a technical requirement but serve as a vital cultural buffer. For these students, the assurance that their trial-and-error processes remain private, protected from the scrutiny of teachers or the judgment of peers, is what specifically triggers the intrinsic motivation to engage deeply with challenging content. These findings suggest that the motivational impact of security may be more pronounced in such high-pressure educational environments compared to more low-stakes or individualistic learning cultures.

Regarding Hypothesis H12, the study reveals that security is a decisive determinant of learning satisfaction, suggesting that students equate data integrity with the overall quality of the educational service. This finding extends the framework of Jo and Park (2023) by highlighting that for digital natives, satisfaction is derived not only from content utility but also from the assurance of operational integrity. In this light, robust security protocols serve as a signal of professionalism and respect for the user, which Terzieva et al. (2025) note effectively enhances the perceived value of the AI tutor. The strong correlation found in this study implies that visible commitment to data protection builds a trust reservoir, allowing learners to overlook minor technical glitches while maintaining high overall satisfaction; conversely, opaque data policies would likely trigger a rapid degradation of user experience (Habbal et al., 2024). This relationship is particularly localized within the Vietnamese educational landscape, where the pace of school digitization has frequently outpaced the development of digital safety literacy and regulatory frameworks. This gap has led to a prevailing skepticism among Vietnamese students and parents regarding the safety of online platforms. Therefore, the high satisfaction levels recorded in this study are a direct reflection of how transparent security measures address a specific societal anxiety. In this context, an AI tutor that demonstrates robust security distinguishes itself from the plethora of unregulated learning apps common in the region, establishing the credibility necessary to sustain long-term user satisfaction. While these design principles are highly effective in Vietnam, their applicability to regions with more mature data protection laws or different technology adoption patterns may vary, suggesting that the “security-satisfaction” link is mediated by the institutional and developmental maturity of the local educational ecosystem.

5.7. Impacts of learning motivation on learning outcomes

The analysis establishes learning motivation as a decisive predictor of learning outcomes, suggesting that in an AI-tutor environment, the student's internal psychological drive is as critical as the technology itself. This finding is strongly corroborated by Duterte et al. (2024), whose research similarly demonstrated that motivational factors are indispensable for enhancing performance in technology-integrated settings. The positive correlation observed in this study also aligns with the conclusions of Chiu et al. (2024), reinforcing the notion that internal drive is requisite for maintaining the cognitive engagement needed to grasp complex STEM concepts. The mechanism behind this result can be further explained through the lens of Yu and Tao (2025), as the data suggests that the motivated students in the sample likely employed the resilience and deep information processing strategies identified in their work. Additionally, given the autonomous nature of using AI tutors, this outcome supports the perspective of Le et al. (2025), confirming that in high-autonomy contexts, self-drive is a primary antecedent of academic success. In the specific context of Vietnam, this result is deeply intertwined with a socio-cultural framework characterized by intense academic competition and the “exam-driven” nature of the National High School Graduation Exam. Within this cultural milieu, learning motivation is often a hybrid of intrinsic interest and extrinsic pressure to achieve social mobility and fulfill familial expectations. The findings suggest that Vietnamese students with high motivation perceive AI tutors not merely as a pedagogical novelty but as a crucial strategic instrument to navigate the limitations of overcrowded physical classrooms and an increasingly demanding STEM curriculum. Furthermore, the technology adoption patterns observed in this study reflect a pragmatic “supplementary” approach common in emerging digital economies; students utilize AI tutors to gain a competitive edge in high-stakes assessments. However, it is essential to establish a boundary for these conclusions: the strength of the relationship between motivation and outcomes may be unique to institutional environments with similar high-pressure characteristics. Consequently, while these findings provide a robust model for the Vietnamese high school demographic, they should not be viewed as universal design principles without considering the cultural norms of the target learners, particularly regarding the degree of institutional pressure and the prevailing societal value placed on academic

performance.

5.8. Impacts of learning satisfaction on learning outcomes

The study confirms a significant positive relationship between learning satisfaction and learning outcomes, indicating that a student's subjective approval of the AI service effectively translates into better academic results. This empirical evidence is consistent with the work of Nia et al. (2023), who established student satisfaction as a reliable indicator of success in online learning environments. However, this relationship is particularly nuanced within the Vietnamese socio-cultural context, where high academic expectations and a result-oriented education system place significant weight on perceived utility and the tangible effectiveness of learning tools. The findings of this study are also reinforced by the frameworks of Yang et al. (2025) and Mekheimer (2025); the data implies that the satisfaction derived from a supportive AI environment likely reduced anxiety, a critical factor in the high-pressure STEM education environment prevalent in Vietnam, and bolstered intrinsic motivation, thereby enabling students to persist with challenging content. Furthermore, this result supports the cognitive resource theory posited by Zhang et al. (2025), suggesting that satisfied learners in this study were able to direct their cognitive resources toward knowledge acquisition rather than grappling with system-induced frustration. Thus, satisfaction is proven here to be a foundational catalyst for learning effectiveness rather than a peripheral emotional state.

Nevertheless, the generalizability of these findings must be interpreted within the specific institutional and cultural boundaries of the Vietnamese educational landscape. In the Vietnamese educational landscape, where traditional pedagogy is often teacher-centered and one-way, the satisfaction gained from the interactive and personalized nature of AI tutors plays a unique role. In a culture that traditionally emphasizes collective classroom instruction and respects teacher authority, the AI tutor provides a vital, low-stakes private space for experimentation and error, which reduces the "fear of losing face" often associated with public questioning. Students who are satisfied with the AI's ability to provide immediate, tailored feedback, a resource often scarce in large Vietnamese classes where student-to-teacher ratios are high, are more likely to engage deeply with the material. This personalized satisfaction effectively bridges the gap between rigid classroom instruction and the student's need for individualized mastery, leading to superior learning outcomes. It is important to acknowledge that these outcomes are situated within a specific pattern of technology adoption where students increasingly view digital tools as necessary supplements to formal schooling to cope with competitive university entrance requirements. Consequently, while the identified psychological pathways provide a robust framework for AI design, these principles may manifest differently in educational contexts with lower academic pressure or more decentralized pedagogical traditions.

6. Implications

6.1. Theoretical implications

This study makes several notable theoretical contributions to the fields of educational technology, human-computer interaction, and STEM pedagogy. By conceptualizing and empirically validating a multidimensional framework of AI tutors, the research extends traditional system success models and advances a more detailed understanding of how specific, designable attributes of AI-driven learning systems shape high school students' learning experiences. Unlike generalized frameworks that focus broadly on system usability or technology acceptance, the proposed model integrates functional, cognitive, and affective dimensions of AI tutors and systematically examines their relationships with learning motivation, learning satisfaction, and measurable learning outcomes. With the integration of the S-O-R paradigm, the study further enriches theory by explaining the psychological pathways through which AI tutors operate. In this study, the AI tutor features function as the stimulus, learning motivation and learning satisfaction represent the organism, and learning outcomes constitute the response. This structure provides a coherent lens through which the internal psychological mechanisms triggered by external system attributes can be theorized and empirically validated.

A primary theoretical contribution lies in the nuanced refinement of the S-O-R framework by identifying a clear hierarchical importance among technical stimuli. While previous research often treats system features as a homogenous group of antecedents, the results reveal that personalization and security act as the primary catalysts for both motivational and evaluative states in the context of high school STEM education. By establishing that these two factors exert the most substantial influence on the organism, this study suggests that psychological safety and individual relevance are the foundational pillars of the S-O-R chain for adolescent learners. This refines existing models by shifting the focus from purely utilitarian or hedonic system attributes toward a more integrated perspective where trust and adaptive pedagogical relevance are viewed as the most potent triggers for internal psychological states.

Furthermore, the study offers a critical refinement of autonomy and learner control within AI-mediated environments. While self-determination theory traditionally positions autonomy as a universal driver of intrinsic motivation, the

application of the S-O-R model demonstrates a clear divergence: control serves as a stimulus that exclusively influences learning satisfaction without activating the motivational state. This suggests that in STEM-focused AI tutoring, control functions as a foundational hygiene factor. It supports a smooth, frustration-free experience (satisfaction) but does not necessarily ignite the internal drive (motivation) required for complex problem-solving. By distinguishing between these two psychological pathways, the study provides a more granular theoretical understanding of how different system attributes map onto specific internal states, moving beyond the confirmation of well-established relationships to offer a specific taxonomy of psychological effects.

Finally, by situating this investigation specifically within high school STEM education, the study deepens theoretical frameworks related to digital pedagogy for adolescent learners. The dual influence of efficiency, enjoyment, and content quality on both motivation and satisfaction indicates that for this demographic, cognitive load reduction and affective engagement are deeply intertwined. The S-O-R model here captures the unique interplay where the complexity of STEM subjects necessitates a stimulus that balances academic rigor with psychological affordances. This validates a robust conceptual basis for future research by positioning motivation and satisfaction not merely as user experience indicators, but as central, distinct mechanisms that translate system design into academic performance, thereby providing a complete theoretical explanation of the internalization process from external stimulus to behavioral response.

6.2. Practical implications

Based on the empirical confirmation of the structural model, this study provides actionable, variable-specific implications for educational technology developers and school administrators seeking to optimize AI tutors in high school STEM education.

Given that personalization and security emerged as the strongest predictors of both learning motivation and satisfaction, these dimensions should be prioritized in system design. To enhance personalization, developers are encouraged to incorporate adaptive learning mechanisms capable of identifying students' knowledge gaps and delivering targeted instructional support. For example, when students repeatedly struggle with specific problem types, AI tutors may provide focused remedial content to reinforce foundational concepts. Furthermore, to address the critical role of security, platforms must transition from opaque terms of service to transparent, user-facing privacy dashboards. Developers should integrate permission-based sharing architectures, allowing high school students to actively control which performance metrics are visible to peers, teachers, or parents, thereby establishing the psychological safety required for deep cognitive engagement.

To leverage the positive effects of efficiency, enjoyment, and learning content, AI tutors should be designed to ensure both technical responsiveness and pedagogical effectiveness. For efficiency, system architectures must prioritize low-latency processing to deliver instantaneous pedagogical feedback, ensuring that students' cognitive flow is not interrupted during complex STEM problem-solving. To translate the enjoyment variable into concrete design, systems should embed sophisticated gamification mechanics, such as interactive 3D simulations for chemistry experiments or logic-based achievement badges, which elevate the hedonic value of the platform. Regarding learning content, it is crucial that AI tutors move beyond providing simple final answers. Systems must be programmed with interactive scaffolding that delivers step-by-step logic breakdowns, guiding students through the intermediate stages of mathematical equations or coding challenges in alignment with standardized school syllabi.

Finally, the findings regarding user control, together with the strong effects of motivation and satisfaction on learning outcomes, suggest important considerations for interface design and instructional integration. Since control significantly enhances learning satisfaction but not motivation, developers should restrict control features to interface customization, such as adjustable reading speeds, visual themes, or flexible navigation menus, while keeping the core pedagogical progression strictly automated to prevent off-task behavior. For educators and administrators, the confirmation that motivation and satisfaction are powerful drivers of learning outcomes means AI tutors should be utilized as real-time diagnostic tools. Dashboards should be provided to teachers that aggregate AI-tracked engagement metrics and frustration indicators, enabling educators to identify disengaged students instantly and apply targeted human mentorship before academic outcomes are negatively impacted.

7. Limitations and recommendations

Despite providing valuable insights into the factors influencing high school STEM students' motivation and satisfaction with AI tutors, this study has several limitations that suggest avenues for future research. First, the cross-sectional design provides only a static snapshot of student perceptions, limiting the ability to examine how engagement with AI tutors evolves over time. Longitudinal studies are therefore recommended to capture the dynamic development of learning motivation, satisfaction, and outcomes. Second, the study's focus on a specific cohort, high school STEM students,

restricts the generalizability of the findings. Future research should consider diverse educational levels and non-STEM subjects to test the applicability of the model across different learning contexts, alongside examining moderating effects of demographic factors such as gender or prior technological experience. Furthermore, the reliance on quantitative survey data may not fully capture the nuanced aspects of the student learning experience; integrating qualitative methodologies would yield a richer understanding. Crucially, a significant limitation of this study is the operationalization of learning outcomes, which relied entirely on students' self-perceived learning gains rather than objective academic metrics. It must be acknowledged that relying on subjective measures introduces interpretative boundaries and raises valid concerns regarding construct validity, as a student's perceived improvement does not necessarily equate to actual academic achievement, particularly within high-stakes STEM disciplines. Because the absence of objective indicators limits the interpretability of the findings regarding actual learning effectiveness, future research must incorporate objective performance data, such as standardized test scores, official grades, or performance-based assessments, to rigorously substantiate and validate the true educational impact of AI tutors. Finally, while the six-dimension framework employed offers a robust foundation, it is not exhaustive; future research could expand the model to include additional factors like pedagogical sophistication. Acknowledging these constraints highlights constructive opportunities for subsequent inquiry, advancing a more comprehensive understanding of how to optimize AI tutors in modern education.

8. Conclusion

This study provides a comprehensive investigation of the dimensions of AI tutors and their influence on high school students in the context of STEM education. Specifically, it explores how six specific AI tutor factors, namely control, efficiency, enjoyment, learning content, personalization, and security, impact learning motivation and learning satisfaction, which in turn drive learning outcomes. Guided by these objectives, a quantitative survey was conducted with 478 high school students who had prior experience using AI-powered STEM tutors. The proposed research model and hypotheses were rigorously tested, generating insights that advance both theoretical understanding and practical strategies in educational technology.

The findings highlight a nuanced and multi-dimensional understanding of AI tutors. A particularly significant finding is the distinct role of control. This dimension was found to be a strong predictor of learning satisfaction, yet it did not significantly influence learning motivation. This suggests that for high school STEM students, the ability to manage the learning pace and process is a foundational expectation for a satisfactory user experience, rather than a feature that actively inspires intrinsic engagement. In contrast, the other five features of AI tutor (efficiency, enjoyment, learning content, personalization, and security) all emerged as powerful predictors of both learning motivation and learning satisfaction. This underscores the need for a holistic approach to service design, where functional, experiential, and affective elements are equally critical. The study further confirmed the strong, positive relationships connecting student attitudes to academic success. Both learning motivation and learning satisfaction were validated as significant drivers of learning outcomes, illustrating the process through which high-quality service translates into tangible educational achievement.

The implications of this study extend significantly into both theoretical and practical domains. Theoretically, this research enriches the literature by applying the S-O-R paradigm to explain the psychological pathways through which AI tutor features influence internal states of learning motivation and learning satisfaction, ultimately leading to learning outcomes. A primary contribution lies in identifying a clear hierarchy among technical stimuli: personalization and security emerge as the most potent catalysts for the "organism" state, suggesting that individual relevance and data protection are the foundational pillars for adolescent learners. This refines existing models by demonstrating that while learner control functions as a foundational "hygiene" factor essential for learning satisfaction, it is the synergistic combination of adaptive pedagogical relevance (personalization) and perceived security that actively drives learning motivation. Practically, these findings offer a concrete blueprint for stakeholders. Developers should prioritize "personalized mastery dashboards" and gap-detection algorithms to operationalize adaptive coaching, alongside transparent "privacy shields" to reinforce the security necessary to support learning motivation and learning satisfaction. Furthermore, technical architectures must prioritize low-latency feedback and interactive scaffolding to maintain cognitive flow. For educators and school administrators, the study suggests that AI tutors should serve as diagnostic tools within the classroom ecosystem, allowing teachers to use AI-generated analytics to identify students with low learning motivation or high levels of dissatisfaction and provide timely human mentorship. By grounding these improvements in verified S-O-R pathways, stakeholders can ensure that technological evolution is logically aligned with the psychological needs of high school students, ultimately leading to more effective and reliable learning outcomes.

In conclusion, as AI-powered systems become increasingly integral to the educational landscape, a deep understanding of their features is essential for harnessing their potential. This study not only addresses this challenge by providing a validated framework but also opens new avenues for future research, such as longitudinal studies and qualitative inquiries to further explore these dynamics. By bridging theoretical gaps in educational technology and offering practical

guidance, this research contributes significantly to advancing the design and application of AI tutors to meet the complex needs of modern high school students.

Declaration of competing interest

The authors affirm that no financial or personal conflicts of interest have influenced the research presented in this paper.

Funding statement

No funding was received for this research.

Acknowledgements

The authors extend their gratitude to the high school students who participated in this survey; their perspectives are the cornerstone of this research. We also acknowledge the valuable contributions of the researchers whose work in this field provided a critical foundation for our study.

References

- Abbas, A., Ekowati, D., Suhariadi, F., & Hamid, S. A. R. (2024). Negative vs. positive psychology: A review of science of well-being. *Integrative Psychological and Behavioral Science*, 58(4), 1091-1122. <https://doi.org/10.1007/s12124-022-09708-1>
- Abisoye, A. (2024). Creating a Conceptual Framework for AI-Powered STEM Education Analytics to Enhance Student Learning Outcomes. *International Journal of Research and Innovation in Social Science*. <https://doi.org/10.54660/IJFMR.2024.5.1.157-167>
- Abreh, M. K., Arthur, F., Akwetey, F. A., & Nortey, S. A. (2025). Modelling STEM students' intention to learn artificial intelligence (AI) in Ghana: a PLS-SEM and fsQCA approach. *Discover Artificial Intelligence*, 5(1), 223. <https://doi.org/10.1007/s44163-025-00466-8>
- Acheampong, R., Balan, T. C., Popovici, D. M., Tuyishime, E., Rekeraho, A., & Voinea, G. D. (2025). Balancing usability, user experience, security and privacy in XR systems: a multidimensional approach: R. Acheampong et al. *International Journal of Information Security*, 24(3), 112. <https://doi.org/10.1007/s10207-025-01025-z>
- Adayilo, D. M., Oyefolahan, I. O., Ndunagu, J. N., Otuya, C., Malcalm, E., & Twabu, K. (2026). AI-Powered Tutoring Systems for Personalised Learning Feedback in Developing Secondary Education Contexts. *Journal of Future Artificial Intelligence and Technologies*, 2(4), 549-564. <https://doi.org/10.62411/faith.3048-3719-290>
- Adel, A. (2024). The convergence of intelligent tutoring, robotics, and IoT in smart education for the transition from industry 4.0 to 5.0. *Smart Cities*, 7(1), 325-369. <https://doi.org/10.3390/smartcities7010014>
- Aggarwal, D., Sharma, D., & Saxena, A. B. (2024). Smart Education: An Emerging Teaching Pedagogy for Interactive and Adaptive Learning Methods. *Journal of Learning and Educational Policy*, 44, 1-9. <https://doi.org/10.55529/jlep.44.1.9>

- Ahmed, S., & Aziz, N. A. (2025). Impact of AI on customer experience in video streaming services: a focus on personalization and trust. *International Journal of Human-Computer Interaction*, 41(12), 7726-7745. <https://doi.org/10.1080/10447318.2024.2400395>
- Akhter, E. (2025). AI In the Classroom: Evaluating The Effectiveness Of Intelligent Tutoring Systems For Multilingual Learners In Secondary Education. *ASRC Procedia: Global Perspectives in Science and Scholarship*, 1(01), 532-563. <https://doi.org/10.63125/gcq1qr39>
- Al-Husari, F., Nakov, O., & Nakov, P. (2025). Multi-Factor Authentication Fatigue: A Growing Concern in User Experience and Security. In *2025 60th International Scientific Conference on Information, Communication and Energy Systems and Technologies (ICEST)* (pp. 1-4). IEEE. <https://doi.org/10.1109/ICEST66328.2025.11098219>
- Ali, R., & Elkot, M. A. (2025). Empowering Educators Integrating Computational Thinking, AI, and STEM: Fostering Innovation Through Technology and Education. In *Empowering STEM Educators With Digital Tools* (pp. 29-58). IGI Global Scientific Publishing. <https://doi.org/10.4018/979-8-3693-9806-7.ch002>
- Al-kfairy, M., Sendaba, O., & Alfandi, O. (2025). Understanding trust in educational Metaverse: the role of social cognitive theory constructs and perceived risks. *Kybernetes*. <https://doi.org/10.1108/K-11-2024-3057>
- Almulla, M. A. (2024). Investigating influencing factors of learning satisfaction in AI ChatGPT for research: University students perspective. *Heliyon*, 10(11). <https://doi.org/10.1016/j.heliyon.2024.e32220>
- Al-Sa'di, A., Yamjal, P., Ahmad, E., Panjabi, R., Allott McPhee, C. A. M., & Guler, O. (2023). Assessing educators' soft skills: Developing a self-assessment instrument. *Administrative Sciences*, 13(9), 208. <https://doi.org/10.3390/admsci13090208>
- Alwakid, W. N., Dahri, N. A., Humayun, M., & Alwakid, G. N. (2025). Integrating AI chatbots for enhancing academic support in business education: A SEM-Based study toward sustainable learning. *The International Journal of Management Education*, 23(3), 101252. <https://doi.org/10.1016/j.ijme.2025.101252>
- Al-Zahrani, A. M., & Alasmari, T. M. (2024). Exploring the impact of artificial intelligence on higher education: The dynamics of ethical, social, and educational implications. *Humanities and Social Sciences Communications*, 11(1), 1-12. <https://doi.org/10.1057/s41599-024-03432-4>
- Anoir, L., Khaldi, M., & Erradi, M. (2024). Intelligent tutor systems and the personalization of pedagogical objects in adaptive e-learning. In *Interactive Media with Next-Gen Technologies and Their Usability Evaluation* (pp. 185-196). Chapman and Hall/CRC. <https://doi.org/10.1201/9781032664828>

- Arsyad, M., Guna, S., & Barus, S. (2024). Enhancing chemistry education through problem-based learning: Analyzing student engagement, motivation, and critical thinking. *International Journal of Curriculum Development, Teaching and Learning Innovation*, 2(3), 110-117. <https://doi.org/10.35335/curriculum.v2i3.178>
- Aunzo, R. T. (2025). Advancing sustainable development goals with educational technology: supporting stem education and fostering innovation through educational technology. In *Advancing sustainable development goals with educational technology* (pp. 65-98). IGI Global Scientific Publishing. <https://doi.org/10.4018/979-8-3693-8242-4.ch003>
- Bai, Y., & Wang, S. (2025). Impact of generative AI interaction and output quality on university students' learning outcomes: a technology-mediated and motivation-driven approach. *Scientific Reports*, 15(1), 24054. <https://doi.org/10.1038/s41598-025-08697-6>
- Balarezo, J. D., Foss, N. J., & Nielsen, B. B. (2024). Organizational learning: Understanding cognitive barriers and what organizations can do about them. *Management Learning*, 55(5), 741-768. <https://doi.org/10.1177/13505076231210635>
- Balaskas, S., Yfantidou, I., Nikolopoulos, T., & Komis, K. (2025). The Psychology of EdTech nudging: persuasion, cognitive load, and intrinsic motivation. *European Journal of Investigation in Health, Psychology and Education*, 15(9), 179. <https://doi.org/10.3390/ejihpe15090179>
- Balo, V. T. M., & Sanchez, J. M. P. (2025). Evaluation of educational assessment module for flexible STEM education. *STEM Education*, 5(1), 130-151. <https://doi.org/10.3934/steme.2025007>
- Bansal, R., Singh, R., Singh, A., Chaudhary, K., & Rasul, T. (2023). *Redefining virtual teaching learning pedagogy*. John Wiley & Sons. <https://doi.org/10.1002/9781119867647>
- Bansal, S., Nangia, P., Chanaliya, N., & Thaichon, P. (2025). The Role of Advanced AI in Improving Chatbot-Driven Consumer Experiences. *Strategic Business Research*, 100023. <https://doi.org/10.1016/j.sbr.2025.100023>
- Bejtíc, Z. (2024). Nurturing Feedback and its Impact on Self-Efficacy, Empowerment, and Professional Growth in Educational and Corporate Environments. *International Journal of Advanced Corporate Learning*, 17(2). <https://doi.org/10.3991/ijac.v17i2.45237>
- Bernardo, A. B., Blanco, E., Tuero, E., & Núñez, J. C. (2025). Exploring the relationship between satisfaction and university dropout: A systematic review. *Review of Education*, 13(3), e70112. <https://doi.org/10.1002/rev3.70112>

- Bhardwaj, V., Zhang, S., Tan, Y. Q., & Pandey, V. (2025). Redefining learning: student-centered strategies for academic and personal growth. In *Frontiers in Education* (Vol. 10, p. 1518602). Frontiers Media SA. <https://doi.org/10.3389/educ.2025.1518602>
- Bognár, L., Ágoston, G., Bacsá-Bán, A., Fauszt, T., Gubán, G., Joós, A., ... & Strauber, G. (2024). Re-evaluating components of classical educational theories in AI-enhanced learning: An empirical study on student engagement. *Education Sciences*, 14(9), 974. <https://doi.org/10.3390/educsci14090974>
- Bouchard, J. (2025). ChatGPT and the separation between knowledge and knower. *Education and Information Technologies*, 30(8), 10091-10110. <https://doi.org/10.1007/s10639-024-13249-y>
- Brusilovsky, P. (2024). AI in education, learner control, and human-AI collaboration. *International Journal of Artificial Intelligence in Education*, 34(1), 122-135. <https://doi.org/10.1007/s40593-023-00356-z>
- Bulathwela, S., Pérez-Ortiz, M., Holloway, C., Cukurova, M., & Shawe-Taylor, J. (2024). Artificial intelligence alone will not democratise education: On educational inequality, techno-solutionism and inclusive tools. *Sustainability*, 16(2), 781. <https://doi.org/10.3390/su16020781>
- Bustamante-Mora, A., Diéguez-Rebolledo, M., Díaz-Arancibia, J., Sánchez-Vázquez, E., & Medina-Gómez, J. (2025). Inclusive Pedagogical Models in STEM: The Importance of Emotional Intelligence, Resilience, and Motivation with a Gender Perspective. *Sustainability*, 17(10), 4437. <https://doi.org/10.3390/su17104437>
- Cheng, Y. M. (2024). What makes learners enhance learning outcomes in MOOCs? Exploring the roles of gamification and personalization. *Interactive Technology and Smart Education*, 21(2), 308-330. <https://doi.org/10.1108/ITSE-05-2023-0097>
- Cheung, G. W., Cooper-Thomas, H. D., Lau, R. S., & Wang, L. C. (2024). Reporting reliability, convergent and discriminant validity with structural equation modeling: A review and best-practice recommendations. *Asia pacific journal of management*, 41(2), 745-783. <https://doi.org/10.1007/s10490-023-09871-y>
- Chisunum, J. I., & Nwadiokwu, C. (2024). Enhancing student engagement through practical production and utilization of instructional materials in an educational technology class: A multifaceted approach. *NIU Journal of Educational Research*, 10(2), 81-89. <https://doi.org/10.58709/niuved.v10i2.2002>
- Chiu, T. K., Moorhouse, B. L., Chai, C. S., & Ismailov, M. (2024). Teacher support and student motivation to learn with Artificial Intelligence (AI) based chatbot. *Interactive Learning Environments*, 32(7), 3240-3256. <https://doi.org/10.1080/10494820.2023.2172044>
- Dai, X., Wang, R. R., Huang, X. F., Wang, X. X., Huang, Y. T., Li, Y., ... & Gao, B. (2024). An analytical model of college students' self-assessed satisfaction with the effectiveness of online learning: a structural equation model

- integrating LICE and SOR models. *Frontiers in Psychology*, 14, 1248729. <https://doi.org/10.3389/fpsyg.2023.1248729>
- Daud, N. M. (2026). Resisting the machine: modeling emotional friction and academic decline in enforced AI learning environments. *Smart Learning Environments*, 13(1), 10. <https://doi.org/10.1186/s40561-026-00435-3>
- Deepshikha, D. (2026). A systematic review on the future of educational assessment: AI-driven grading and personalised feedback in higher education. In *Artificial Intelligence in Education* (Vol. 2, No. 2, pp. 75-115). Emerald Publishing Limited. <https://doi.org/10.1108/AIIE-03-2025-0036>
- Deng, R., Jiang, M., Yu, X., Lu, Y., & Liu, S. (2025). Does ChatGPT enhance student learning? A systematic review and meta-analysis of experimental studies. *Computers & Education*, 227, 105224. <https://doi.org/10.1016/j.compedu.2024.105224>
- Ding, Z., & Xue, W. (2025). Navigating anxiety in digital learning: How AI-driven personalization and emotion recognition shape EFL students' engagement. *Acta Psychologica*, 260, 105466. <https://doi.org/10.1016/j.actpsy.2025.105466>
- Du, H. S., Chen, T., & Ke, X. (2025). Personalized or Pressurized? The Influence of AI-Powered Personalized Recommendation Triggered Information Stressor on Consumers' Discontinuance Purchase Behavior. *International Journal of Human-Computer Interaction*, 1-21. <https://doi.org/10.1080/10447318.2025.2598452>
- Dunne, G. (2025). Resisting Edutainment: How Reward Deficiency Syndrome Fuels the Student Attention Crisis. *Studies in Philosophy and Education*, 1-27. <https://doi.org/10.1007/s11217-025-10022-7>
- Duong, N. T., Pham Thi, T. D., & Pham, V. K. (2025). A comparative study on AI-based learning behaviors: evidence from Vietnam. *International Journal of Human-Computer Interaction*, 41(16), 10007-10023. <https://doi.org/10.1080/10447318.2024.2430433>
- Duterte, J. P. (2024). Technology-enhanced learning environments: Improving engagement and learning. *International Journal of Research and Innovation in Social Science*, 8(10), 1305-131. <https://doi.org/10.47772/IJRISS.2024.8100111>
- Erensoy, A., Mathrani, A., Schnack, A., Elms, J., & Baghaei, N. (2024). Consumer behavior in immersive virtual reality retail environments: A systematic literature review using the stimuli-organisms-responses (S-O-r) model. *Journal of Consumer Behaviour*, 23(6), 2781-2811. <https://doi.org/10.1002/cb.2374>

- Estipona, E. P., & Santos, M. S. M. D. (2025). Linking wellbeing to success: life satisfaction on mathematics performance of Philippine public high school students. In *Frontiers in Education* (Vol. 10, p. 1540813). Frontiers Media SA. <https://doi.org/10.3389/feduc.2025.1540813>
- Fan, J., & Zhang, Q. (2024). From literacy to learning: The sequential mediation of attitudes and enjoyment in AI-assisted EFL education. *Heliyon*, 10(17). <https://doi.org/10.1016/j.heliyon.2024.e37158>
- Farhan, N. D., Sadiq, B. H., Zwayyer, M. H., & Arnout, B. A. (2024). The impact of using artificial intelligence techniques in improving the quality of educational services/case study at the University of Baghdad. In *Frontiers in Education* (Vol. 9, p. 1474370). Frontiers Media SA. <https://doi.org/10.3389/feduc.2024.1474370>
- Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of marketing research*, 18(1), 39-50. <https://doi.org/10.1177/002224378101800104>
- Fu, C. J., Silalahi, A. D. K., Shih, I. T., Phuong, D. T. T., Eunike, I. J., & Jargalsaikhan, S. (2024). Assessing ChatGPT's information quality through the lens of user information satisfaction and information quality theory in higher education: A theoretical framework. *Human Behavior and Emerging Technologies*, 2024(1), 8114315. <https://doi.org/10.1155/2024/8114315>
- Gaskin, J. E., Lowry, P. B., Rosengren, W., & Fife, P. T. (2025). Essential validation criteria for rigorous covariance-based structural equation modelling. *Information Systems Journal*, 35(6), 1630-1661. <https://doi.org/10.1111/isj.12598>
- Gavrilas, L., & Kotsis, K. T. (2025). Integrating learning theories and innovative pedagogies in STEM education: A comprehensive review. *Eurasian Journal of Science and Environmental Education*, 5(1), 11-17. <https://doi.org/10.30935/ejsee/16538>
- Gkintoni, E., Antonopoulou, H., Sortwell, A., & Halkiopoulos, C. (2025). Challenging cognitive load theory: The role of educational neuroscience and artificial intelligence in redefining learning efficacy. *Brain Sciences*, 15(2), 203. <https://doi.org/10.3390/brainsci15020203>
- Glick, D., Miedijensky, S., & Zhang, H. (2024). Examining the effect of AI-powered virtual-human training on STEM majors' self-regulated learning behavior. In *Frontiers in Education* (Vol. 9, p. 1465207). Frontiers Media SA. <https://doi.org/10.3389/feduc.2024.1465207>
- Gligorea, I., Cioca, M., Oancea, R., Gorski, A. T., Gorski, H., & Tudorache, P. (2023). Adaptive learning using artificial intelligence in e-learning: A literature review. *Education Sciences*, 13(12), 1216. <https://doi.org/10.3390/educsci13121216>

- Gomes, D. (2025). A Comprehensive Study of Advancements in Intelligent Tutoring Systems Through Artificial Intelligent Education Platforms. In *Improving Student Assessment With Emerging AI Tools* (pp. 213-244). IGI Global Scientific Publishing. <https://doi.org/10.4018/979-8-3693-6170-2.ch008>
- Goss, H. (2022). Student learning outcomes assessment in higher education and in academic libraries: A review of the literature. *The Journal of Academic Librarianship*, 48(2), 102485. <https://doi.org/10.1016/j.acalib.2021.102485>
- Guo, W., Wang, J., Li, N., & Wang, L. (2025). The impact of teacher emotional support on learning engagement among college students mediated by academic self-efficacy and academic resilience. *Scientific Reports*, 15(1), 3670. <https://doi.org/10.1038/s41598-025-88187-x>
- Gyasi, J. F., Zheng, L., & Zhou, Y. (2021). Perusing the past to propel the future: A systematic review of STEM learning activity based on activity theory. *Sustainability*, 13(16), 8828. <https://doi.org/10.3390/su13168828>
- Gyonyoru, K. I. K., & Katona, J. (2024). Student perceptions of AI-enhanced adaptive learning systems: A pilot survey. In *2024 IEEE 7th International Conference and Workshop Óbuda on Electrical and Power Engineering (CANDO-EPE)* (pp. 93-98). IEEE. <https://doi.org/10.1109/CANDO-EPE65072.2024.10772884>
- Gyönyörű, K. I. K., & Katona, J. (2025). Comprehensive overview of the concept and applications of AI-based adaptive learning. *Acta Polytechnica Hungarica*. <https://doi.org/10.12700/APH.22.3.2025.3.9>
- Habbal, A., Ali, M. K., & Abuzaraida, M. A. (2024). Artificial Intelligence Trust, risk and security management (AI trism): Frameworks, applications, challenges and future research directions. *Expert Systems with Applications*, 240, 122442. <https://doi.org/10.1016/j.eswa.2023.122442>
- Hair, J., & Alamer, A. (2022). Partial Least Squares Structural Equation Modeling (PLS-SEM) in second language and education research: Guidelines using an applied example. *Research Methods in Applied Linguistics*, 1(3), 100027. <https://doi.org/10.1016/j.rmal.2022.100027>
- Hair, J., Risher, J. J., Sarstedt, M., & Ringle, C. M. (2019). When to use and how to report the results of PLS-SEM. *European business review*, 31(1), 2-24. <https://doi.org/10.1108/EBR-11-2018-0203>
- Hang, N. N. T., Ngoc, M. N., & Van, H. N. (2024). Current situation and solutions to improve the quality of teaching in STEM education in general schools: a case study in some northern provinces of Vietnam. *European Journal of Education Studies*, 11(1). <http://dx.doi.org/10.46827/ejes.v11i1.5172>
- Hernández-Campos, M., Gonzalez-Torres, A., & García-Peñalvo, F. J. (2025). Learning outcomes evaluation through learning analytics systems in higher education: A systematic literature review. *SAGE Open*, 15(3), 21582440251347374. <https://doi.org/10.1177/21582440251347374>

- Ho, I. M. K., Cheong, K. Y., & Weldon, A. (2021). Predicting student satisfaction of emergency remote learning in higher education during COVID-19 using machine learning techniques. *Plos one*, 16(4), e0249423. <https://doi.org/10.1371/journal.pone.0249423>
- Huang, J., Zhong, Y., & Chen, X. (2025). Adaptive and personalized learning in STEM education using high-performance computing and artificial intelligence. *The Journal of Supercomputing*, 81(8), 981. <https://doi.org/10.1007/s11227-025-07481-7>
- Hulland, J., Baumgartner, H., & Smith, K. M. (2018). Marketing survey research best practices: evidence and recommendations from a review of JAMS articles. *Journal of the academy of marketing science*, 46(1), 92-108. <https://doi.org/10.1007/s11747-017-0532-y>
- Iglesias-Rodriguez, A., Hernandez-Martin, A., Martin-Gonzalez, Y., & Herraiz-Corredera, P. (2021). Design, validation and implementation of a questionnaire to assess teenagers' digital competence in the area of communication in digital environments. *Sustainability*, 13(12), 6733. <https://doi.org/10.3390/su13126733>
- Imeri, F., Rushiti, S., Sadiki, F., Memeti, A., & Idrizi, F. (2024). Education in STEM and High School Students' Views, Career Outlooks, and Interest in IT Fields. In *2024 13th Mediterranean Conference on Embedded Computing (MECO)* (pp. 1-5). IEEE. <https://doi.org/10.1109/MECO62516.2024.10577914>
- Jannini, A. V. S., Akdemir, Z., & Menekse, M. (2024). Achievement goal theory in STEM education: A systematic review. *Journal of Engineering Education*, 113(4), 986-1007. <https://doi.org/10.1002/jee.20585>
- Jauhiainen, J. S., & Guerra, A. G. (2024). Generative AI and education: dynamic personalization of pupils' school learning material with ChatGPT. In *Frontiers in Education* (Vol. 9, p. 1288723). Frontiers Media SA. <https://doi.org/10.3389/feduc.2024.1288723>
- Jeilani, A., & Abubakar, S. (2025). Perceived institutional support and its effects on student perceptions of AI learning in higher education: the role of mediating perceived learning outcomes and moderating technology self-efficacy. In *Frontiers in Education* (Vol. 10, p. 1548900). Frontiers Media SA. <https://doi.org/10.3389/feduc.2025.1548900>
- Ji, H., Suo, L., & Chen, H. (2024). AI performance assessment in blended learning: mechanisms and effects on students' continuous learning motivation. *Frontiers in Psychology*, 15, 1447680. <https://doi.org/10.3389/fpsyg.2024.1447680>
- Jiang, S., Zhang, Z., Xu, H., & Pan, Y. (2024). What influences users' continuous behavioral intention in cultural heritage virtual tourism: Integrating experience economy theory and stimulus–organism–response (SOR) model. *Sustainability*, 16(23), 10231. <https://doi.org/10.3390/su162310231>

- Jo, H., & Park, D. H. (2023). Mechanisms for successful management of enterprise resource planning from user information processing and system quality perspective. *Scientific Reports*, 13(1), 12678. <https://doi.org/10.1038/s41598-023-39787-y>
- Johnpaul, M., Miryala, R. S. B., Jayaprakashnarayana, G., & Miryala, R. K. (2025). Pros and cons of artificial intelligence on metacognition: A myopic state with long-term consequences on human learning. In *Hybrid and advanced technologies* (pp. 98-103). CRC Press. <https://doi.org/10.1201/9781003559139>
- Kassenkhan, A. M., Moldagulova, A. N., & Serbin, V. V. (2025). Gamification and Artificial Intelligence in Education: A Review of Innovative Approaches to Fostering Critical Thinking. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2025.3576147>
- Kausar, F., & Kumar, J. P. (2025). Bridging the Gap: Generative AI for Personalized Learning and Tutoring. In *International Symposium on Generative AI and Education* (pp. 153-161). Cham: Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-98476-1_13
- Kayyali, M. (2025). The Impact of Personalized Learning on Student Engagement and Achievement in STEAM. In *Integrating Personalized Learning Methods Into STEAM Education* (pp. 51-78). IGI Global Scientific Publishing. <https://doi.org/10.4018/979-8-3693-7718-5.ch003>
- Kestin, G., Miller, K., Klaes, A., Milbourne, T., & Ponti, G. (2025). AI tutoring outperforms in-class active learning: an RCT introducing a novel research-based design in an authentic educational setting. *Scientific Reports*, 15(1), 17458. <https://doi.org/10.1038/s41598-025-97652-6>
- Kexin, Z., & Teo, P. C. (2023). The adoption of Stimulus-Organism-Response (SOR) model in the social commerce literature. *International Journal of Academic Research in Business and Social Sciences*, 13(7), 1879-1886. <https://doi.org/10.6007/IJARBS/v13-i7/17290>
- Khairuddin, Z., Shahabani, N. S., Ahmad, S. N., Ahmad, A. R., & Zamri, N. A. (2024). Students' perceptions on the artificial intelligence (AI) tools as academic support. *Malaysian Journal of Social Sciences and Humanities (MJSSH)*, 9(11), e003087-e003087. <https://doi.org/10.47405/mjssh.v9i11.3087>
- Khanal, B., & Chhetri, D. B. (2024). A pilot study approach to assessing the reliability and validity of relevancy and efficacy survey scale. *Janabhawana Research Journal*, 3(1), 35-49. <https://doi.org/10.3126/jrj.v3i1.68384>
- Khoso, A. K., Honggang, W., & Darazi, M. A. (2025). Trust and attitude towards AI as pathways to creativity: a TAM Model study of EFL students' digital literacy and AI acceptance. *Humanities and Social Sciences Communications*, 13(1), 69. <https://doi.org/10.1057/s41599-025-06362-x>

- Kim, K., Hwang, J., Zo, H., & Lee, H. (2014). Understanding users' continuance intention toward smartphone augmented reality applications. *Information Development*, 32(2), 161–174. <https://doi.org/10.1177/0266666914535119>
- Kim, M. K., Jhee, S. Y., & Han, S. L. (2025). The Impact of Chat GPT's Quality Factors on~ Perceived Usefulness, Perceived Enjoyment, and~ Continuous Usage Intention Using the IS Success Model. *Asia Marketing Journal*, 26(4), 243-254. <https://doi.org/10.53728/2765-6500.1641>
- Kim, Y. E., Brady, A. C., & Wolters, C. A. (2018). Development and validation of the brief regulation of motivation scale. *Learning and Individual Differences*, 67, 259-265. <https://doi.org/10.1016/j.lindif.2017.12.010>
- Klein, C. R., & Klein, R. (2025). The extended hollowed mind: why foundational knowledge is indispensable in the age of AI. *Frontiers in Artificial Intelligence*, 8, 1719019. <https://doi.org/10.3389/frai.2025.1719019>
- Kock, N. (2015). Common method bias in PLS-SEM: A full collinearity assessment approach. *International Journal of e-Collaboration (ijec)*, 11(4), 1-10. <https://doi.org/10.4018/ijec.2015100101>
- Kumar, P., Saxena, C., & Baber, H. (2021). Learner-content interaction in e-learning-the moderating role of perceived harm of COVID-19 in assessing the satisfaction of learners. *Smart Learning Environments*, 8(1), 5. <https://doi.org/10.1186/s40561-021-00149-8>
- Le, H. V. (2025). Assessing student satisfaction with MOOCs: a comprehensive analysis of Coursera's instructional design and learner experience. *The International Journal of Information and Learning Technology*, 42(2), 207-223. <https://doi.org/10.1108/IJILT-07-2024-0159>
- Le, H., Shen, Y., Li, Z., Xia, M., Tang, L., Li, X., ... & Fan, Y. (2025). Breaking human dominance: Investigating learners' preferences for learning feedback from generative AI and human tutors. *British Journal of Educational Technology*. <https://doi.org/10.1111/bjet.13614>
- Li, W., Zhang, X., Li, J., Yang, X., Li, D., & Liu, Y. (2024). An explanatory study of factors influencing engagement in AI education at the K-12 Level: an extension of the classic TAM model. *Scientific Reports*, 14(1), 13922. <https://doi.org/10.1038/s41598-024-64363-3>
- Liang, Y. D., Chen, S., Abeysekera, R., O'Sullivan, H., Bray, J., & Keevill-Savage, I. (2024). Examining the adoption of technology-enhanced learning in universities and its effects on student performance, satisfaction, and motivation. *Computers and Education Open*, 7, 100223. <https://doi.org/10.1016/j.caeo.2024.100223>
- Lin, H., & Chen, Q. (2024). Artificial intelligence (AI)-integrated educational applications and college students' creativity and academic emotions: students and teachers' perceptions and attitudes. *BMC psychology*, 12(1), 487. <https://doi.org/10.1186/s40359-024-01979-0>

- Lin, Z., Dai, Y., & Ng, O. L. (2025). Constructionism in K-12 AI Literacy Education: A Systematic Review of Pedagogical Designs, Student Outcomes, and Learning Mechanisms. *Journal of Educational Computing Research*, 07356331251360442. <https://doi.org/10.1177/07356331251360442>
- Lin-Siegler, X., Lovett, B. J., Du, Y., Yamane, K., Wang, K., & Hadis, S. (2023). What experiences constitute failures? High school students' reflections on their struggles in STEM classes. *Annals of the New York Academy of Sciences*, 1524(1), 105-117. <https://doi.org/10.1111/nyas.14990>
- Liu, C., Wang, G. C., & Wang, H. F. (2025). The application of artificial intelligence in engineering education: A systematic review. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2025.3532595>
- Liu, J., Sun, D., Sun, J., Wang, J., & Yu, P. L. H. (2025). Designing a generative AI enabled learning environment for mathematics word problem solving in primary schools: Learning performance, attitudes and interaction. *Computers and Education: Artificial Intelligence*, 9, 100438. <https://doi.org/10.1016/j.caeai.2025.100438>
- Liu, Y., & Liu, Y. (2025). Advancing STEM Education for Sustainability: The Impact of Graphical Knowledge Visualization and User Experience on Continuance Intention in Mixed-Reality Environments. *Sustainability*, 17(9), 3869. <https://doi.org/10.3390/su17093869>
- Maia, D. M., Dos Santos, S. C., Lima, L. G., Franca, V. L., Lima, A. H., & Andrade, D. (2024). Critical factors for a reliable AI in tutoring systems on accuracy, effectiveness, and responsibility. In *2024 IEEE Frontiers in Education Conference (FIE)* (pp. 1-9). IEEE. <https://doi.org/10.1109/FIE61694.2024.10893361>
- Maier, U., & Klotz, C. (2022). Personalized feedback in digital learning environments: Classification framework and literature review. *Computers and education: Artificial intelligence*, 3, 100080. <https://doi.org/10.1016/j.caeai.2022.100080>
- Marengo, A., & Pange, P. J. (2024). Envisioning general AI in higher education: transforming learning paradigms and pedagogies. In *The Learning Ideas Conference* (pp. 330-344). Cham: Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-72430-5_28
- Martin, F., & Bolliger, D. U. (2022). Developing an online learner satisfaction framework in higher education through a systematic review of research. *International Journal of Educational Technology in Higher Education*, 19(1), 50. <https://doi.org/10.1186/s41239-022-00355-5>
- McKay, B., Bacelar, M. F., Parma, J. O., Miller, M. W., & Carter, M. J. (2025). The combination of reporting bias and underpowered study designs has substantially exaggerated the motor learning benefits of self-controlled practice and enhanced expectancies: A meta-analysis. *International Review of Sport and Exercise Psychology*, 18(1), 242-262. <https://doi.org/10.1080/1750984X.2023.2207255>

- Mehrabian, A., & Russell, J. A. (1974). The basic emotional impact of environments. *Perceptual and motor skills*, 38(1), 283-301. <https://doi.org/10.2466/pms.1974.38.1.283>
- Mekheimer, M. (2025). Technological self-efficacy, motivation, and contextual factors in advanced EFL e-learning: a mixed-methods study of strategy use and satisfaction. *Humanities and Social Sciences Communications*, 12(1), 1-18. <https://doi.org/10.1057/s41599-025-04947-0>
- Melo, N. (2024). The Impact of AI-Powered Tutoring Platforms on Student Achievement in STEM Subjects: The Case of Cameroon. Available at SSRN 5249691. <https://dx.doi.org/10.2139/ssrn.5249691>
- Micatka, N. K. (2025). Learning to avoid: The long-term effects of adolescent welfare participation on voting habits in adulthood. *Policy studies journal*, 53(4), 1065-1087. <https://doi.org/10.1111/psj.70003>
- Mierzwa-Kamińska, E. (2025). Positive and Negative Emotions: Theoretical Considerations. In *Foreign Language Enjoyment and Anxiety Among Polish EFL Students: Theory and Practical Implications for Foreign Language Educators* (pp. 7-58). Cham: Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-86122-2_2
- Mikić, V., Keković, G., Mangaroska, K., Ilić, M., Kopanja, L., & Vesin, B. (2025). Exploring learner engagement in e-learning environments: A predictive analytics perspective. *International Journal of Human-Computer Interaction*, 1-24. <https://doi.org/10.1080/10447318.2025.2535530>
- Mohamed, A. M., Shaaban, T. S., Bakry, S. H., Guillén-Gámez, F. D., & Strzelecki, A. (2025). Empowering the faculty of education students: Applying AI's potential for motivating and enhancing learning. *Innovative Higher Education*, 50(2), 587-609. <https://doi.org/10.1007/s10755-024-09747-z>
- Moors, A. (2025). Emotions as high-impact decisions: A goal-directed theory. *Emotion Review*, 17540739261422553. <https://doi.org/10.1177/17540739261422553>
- Nagaraj, B. K., Kalaivani, A., Begum, S., Akila, S., & Sachdev, H. K. (2023). The emerging role of artificial intelligence in stem higher education: A critical review. *International Research Journal of Multidisciplinary Technovation*, 5(5), 1-19. <https://doi.org/10.54392/irjmt2351>
- Nalluri, V., Chen, L. S., Huynh-Cam, T. T., & Appana, S. M. (2024). Assessing Service Quality and E-Satisfaction of E-Learning Platforms: A TISM-Based SERVQUAL Model Approach. In *2024 6th International Workshop on Artificial Intelligence and Education (WAIE)* (pp. 260-264). IEEE. <https://doi.org/10.1109/WAIE63876.2024.00054>
- Ngu, B. H., & Phan, H. P. (2024). Instructional approach and acquisition of mathematical proficiency: Theoretical insights from learning by comparison and cognitive load theory. *Asian Journal for Mathematics Education*, 3(3), 357-379. <https://doi.org/10.1177/27527263241266765>

- Nguyen, M. H., Jin, R., Hoang, G., Nguyen, M. H. T., Nguyen, P. L., Le, T. T., ... & Vuong, Q. H. (2023). Examining contributors to Vietnamese high school students' digital creativity under the serendipity-mindsponge-3D knowledge management framework. *Thinking Skills and Creativity*, 49, 101350. <https://doi.org/10.1016/j.tsc.2023.101350>
- Nguyen, P. L. (2024). Vietnam's STEM Education Landscape: Evolution, Challenges, and Policy Interventions. *Vietnam Journal of Education*, 177-189. <https://doi.org/10.52296/vje.2024.389>
- Nguyen, T. T. H. (2025). The impact of translation on language learning-an insight from the functional approach to translation. *VNU Journal of Foreign Studies*, 41(1S (Special Issue)), 104-119. <https://doi.org/10.63023/2525-2445/jfs.ulis.5404>
- Nguyen, V. B., & Nguyen, V. B. (2025). Gamification in Vietnamese education: Assessing psychological need satisfaction, intrinsic motivation, and learning effectiveness. *Learning and Motivation*, 89, 102101. <https://doi.org/10.1016/j.lmot.2025.102101>
- Nia, H. S., Marôco, J., She, L., Khoshnavay Fomani, F., Rahmatpour, P., Stepanovic Ilic, I., ... & Reardon, J. (2023). Student satisfaction and academic efficacy during online learning with the mediating effect of student engagement: A multi-country study. *PLoS One*, 18(10), e0285315. <https://doi.org/10.1371/journal.pone.0285315>
- Ogunmokun, O. A., Unverdi-Creig, G. I., Said, H., Avci, T., & Eluwole, K. K. (2021). Consumer well-being through engagement and innovation in higher education: A conceptual model and research propositions. *Journal of Public Affairs*, 21(1), e2100. <https://doi.org/10.1002/pa.2100>
- Oliso, Z. Z., Alemu, D. D., & Jansen, J. D. (2024). The impact of educational service quality on student academic performance in Ethiopian public universities: a mediating role of students' satisfaction. *Journal of International Education in Business*, 17(2), 340-370. <https://doi.org/10.1108/JIEB-07-2023-0042>
- Ortiz-Rojas, M., Chiluiza, K., Valcke, M., & Bolanos-Mendoza, C. (2025). How gamification boosts learning in STEM higher education: a mixed methods study. *International Journal of STEM Education*, 12(1), 1. <https://doi.org/10.1186/s40594-024-00521-3>
- Osei, P. C., & Bjorklund, D. F. (2024). Motivating the learning process: integrating self-determination theory into a dynamical systems framework. *Educational Psychology Review*, 36(3), 89. <https://doi.org/10.1007/s10648-024-09934-6>

- Ouyang, F., Zheng, L., & Jiao, P. (2022). Artificial intelligence in online higher education: A systematic review of empirical research from 2011 to 2020. *Education and Information Technologies*, 27(6), 7893-7925. <https://doi.org/10.1007/s10639-022-10925-9>
- Pan, J., Ishak, N. A., & Qin, Y. (2024). The application of Moore's online learning interactions model in learning outcomes: The SOR (stimulus-organism-response) paradigm perspective. *Heliyon*, 10(7). <https://doi.org/10.1016/j.heliyon.2024.e28505>
- Pandita, A., & Kiran, R. (2023). The technology interface and student engagement are significant stimuli in sustainable student satisfaction. *Sustainability*, 15(10), 7923. <https://doi.org/10.3390/su15107923>
- Panditharathna, R., Liu, Y., de Macedo Bergamo, F. V., Appiah, D., Trim, P. R., & Lee, Y. I. (2024). How Cyber Security Enhances Trust and Commitment to Customer Retention: The Mediating Role of Robotic Service Quality. *Big Data and Cognitive Computing*, 8(11), 165. <https://doi.org/10.3390/bdcc8110165>
- Parviz, M. (2024). AI in education: Comparative perspectives from STEM and Non-STEM instructors. *Computers and Education Open*, 6, 100190. <https://doi.org/10.1016/j.caeo.2024.100190>
- Paul, S. K. (2025). Over-efforting as weakness of will. *Synthese*, 206(6), 291. <https://doi.org/10.1007/s11229-025-05391-y>
- Pellas, N. (2025). The impact of AI-generated instructional videos on problem-based learning in science teacher education. *Education Sciences*, 15(1), 102. <https://doi.org/10.3390/educsci15010102>
- Pellas, N. (2026). Effects of Generative AI Feedback and Interactive Video Assessment on Student Learning Achievement in Philological Content Creation Courses. *Journal of Educational Computing Research*, 64(1), 16-58. <https://doi.org/10.1177/07356331251372800>
- Pham, M., Nguyen, P. X. T., & Trinh, N. N. (2025). Exploring emotional and technological drivers of e-learning continuance intention among university students. *Discover Education*, 4(1), 1-22. <https://doi.org/10.1007/s44217-025-00800-3>
- Pham, T. C., & Ngo, T. T. A. (2026). AI Tutor-Based Language Learning: Linking Service Quality to Learners' Continuance Intention Through A Dual-Pathway Model. *Journal of Information Technology Education: Research*, 25, 07. <https://doi.org/10.28945/5720>
- Popescu, M., Baruh, L., & Sudhakar, S. (2024). Role-based privacy cynicism and local privacy activism: How data stewards navigate privacy in higher education. *Big Data & Society*, 11(2), 20539517241240664. <https://doi.org/10.1177/20539517241240664>

- Qin, F., Hao, Z., Yu, J., Liu, Z., & Zhang, Y. (2025). AI instructional agent improves student's perceived learner control and learning outcome: empirical evidence from a randomized controlled trial. *arXiv preprint arXiv:2505.22526*.
<https://doi.org/10.48550/arXiv.2505.22526>
- Rad, H. S. (2026). Exploring false growth mindsets among second language learners: implications for motivation and achievement. *Educational and Developmental Psychologist*, 43(1), 70-90.
<https://doi.org/10.1080/20590776.2025.2545255>
- Rahman, M. S., Sabbir, M. M., Zhang, J., Moral, I. H., & Hossain, G. M. S. (2023). Examining students' intention to use ChatGPT: Does trust matter?. *Australasian Journal of Educational Technology*, 39(6), 51-71.
<https://doi.org/10.14742/ajet.8956>
- Rajabalee, Y. B., & Santally, M. I. (2021). Learner satisfaction, engagement and performances in an online module: Implications for institutional e-learning policy. *Education and information technologies*, 26(3), 2623-2656.
<https://doi.org/10.1007/s10639-020-10375-1>
- Rana, S., & Chicone, R. (2025). Gamification and immersive learning with AI. In *Fortifying the future: harnessing AI for transformative cybersecurity training* (pp. 51-75). Cham: Springer Nature Switzerland.
https://doi.org/10.1007/978-3-031-81780-9_3
- Rasyid, A., Rinto, R., & Susanti, M. (2023). Project-based learning through the STEM approach in elementary schools: How to improve problem-solving ability. *Journal of Education For Sustainable Innovation*, 1(1), 1-8.
<https://doi.org/10.56916/jesi.v1i1.477>
- Razak, M. B. A., & Khan, M. R. B. (2024). Student Perception of AI-Powered Service Quality and Customer Satisfaction: A Case Study of Higher Learning Institution in Malaysia. *Malaysian Journal of Business, Economics and Management*, 183-195. <https://doi.org/10.56532/mjbem.v3i2.82>
- Reid, L., Button, D., & Brommeyer, M. (2023). Challenging the myth of the digital native: A narrative review. *Nursing Reports*, 13(2), 573-600. <https://doi.org/10.3390/nursrep13020052>
- Rizki, I. A., & Suprpto, N. (2024). Project-oriented problem-based learning through SR-STEM to foster students' critical thinking skills in renewable energy material. *Journal of Science Education and Technology*, 33(4), 526-541. <https://doi.org/10.1007/s10956-024-10102-2>
- Roemer, E., Schubert, F., & Henseler, J. (2021). HTMT2—an improved criterion for assessing discriminant validity in structural equation modeling. *Industrial management & data systems*, 121(12), 2637-2650.
<https://doi.org/10.1108/IMDS-02-2021-0082>

- Rohles, B., Backes, S., Fischbach, A., Amadiou, F., & Koenig, V. (2022). Creating positive learning experiences with technology: A field study on the effects of user experience for digital concept mapping. *Heliyon*, 8(4). <https://doi.org/10.1016/j.heliyon.2022.e09246>
- Ruijia, Z., Wenling, L., & Xuemei, Z. (2025). The impact of Information and Communication Technology (ICT) on learning outcomes in early childhood and primary education: a meta-analysis of moderating factors. *Frontiers in Psychology*, 16, 1540169. <https://doi.org/10.3389/fpsyg.2025.1540169>
- Ruiz, W. D., & Yabut, H. J. (2024). Autonomy and identity: the role of two developmental tasks on adolescent's wellbeing. *Frontiers in Psychology*, 15, 1309690. <https://doi.org/10.3389/fpsyg.2024.1309690>
- Sajja, R., Sermet, Y., Fodale, B., & Demir, I. (2026). Evaluating AI-powered learning assistants in engineering higher education with implications for student engagement, ethics, and policy. *Scientific Reports*. <https://doi.org/10.1038/s41598-026-39237-5>
- Sarstedt, M., Hair, J. F., Pick, M., Liengaard, B. D., Radomir, L., & Ringle, C. M. (2022). Progress in partial least squares structural equation modeling use in marketing research in the last decade. *Psychology & Marketing*, 39(5), 1035-1064. <https://doi.org/10.1002/mar.21640>
- Seo, Y. J., & Um, K. H. (2023). The role of service quality in fostering different types of perceived value for student blended learning satisfaction. *Journal of Computing in Higher Education*, 35(3), 521-549. <https://doi.org/10.1007/s12528-022-09336-z>
- Sharma, K., Nguyen, A., & Hong, Y. (2024). Self-regulation and shared regulation in collaborative learning in adaptive digital learning environments: A systematic review of empirical studies. *British Journal of Educational Technology*, 55(4), 1398-1436. <https://doi.org/10.1111/bjet.13459>
- Sharma, P. N., Shmueli, G., Sarstedt, M., Danks, N., & Ray, S. (2021). Prediction-oriented model selection in partial least squares path modeling. *Decision Sciences*, 52(3), 567-607. <https://doi.org/10.1111/deci.12329>
- Shi, L., Ding, A. C., & Choi, I. (2024). Investigating teachers' use of an AI-enabled system and their perceptions of AI integration in science classrooms: A case study. *Education Sciences*, 14(11), 1187. <https://doi.org/10.3390/educsci14111187>
- Shin, D. (2023). Embodying algorithms, enactive artificial intelligence and the extended cognition: You can see as much as you know about algorithm. *Journal of Information Science*, 49(1), 18-31. <https://doi.org/10.1177/0165551520985495>

- Sobaih, A. E. E., Hasanein, A., & Elshaer, I. A. (2022). Higher education in and after COVID-19: The impact of using social network applications for e-learning on students' academic performance. *Sustainability*, 14(9), 5195. <https://doi.org/10.3390/su14095195>
- Sofologi, M., Katsarou, D. V., Tsirides, A., & Efthymiou, E. (2025). The synergy of artificial intelligence and education: New perspectives of an innovative artificial tutoring in school settings. In *Prompt Engineering and Generative AI Applications for Teaching and Learning* (pp. 189-202). IGI Global Scientific Publishing. <https://doi.org/10.4018/979-8-3693-7332-3.ch012>
- Stanislaw, P. (2024). Solving the global STEM educational crisis using Cognitive Load Optimization and Artificial Intelligence—A preliminary comparative analysis. *Eurasia Journal of Mathematics, Science and Technology Education*, 20(5). <https://doi.org/10.29333/ejmste/14448>
- Stratton, S. J. (2023). Population sampling: Probability and non-probability techniques. *Prehospital and Disaster Medicine*, 38(2), 147-148. <https://doi.org/10.1017/S1049023X23000304>
- Sumi, R. S., & Kabir, G. (2021). Satisfaction of e-learners with electronic learning service quality using the servqual model. *Journal of Open Innovation: Technology, Market, and Complexity*, 7(4), 227. <https://doi.org/10.3390/joitmc7040227>
- Sun, D., Cheng, G., Yu, P. L. H., Jia, J., Zheng, Z., & Chen, A. (2025). Personalized stem education empowered by artificial intelligence: a comprehensive review and content analysis. *Interactive Learning Environments*, 1-23. <https://doi.org/10.1080/10494820.2025.2462156>
- Tan, L. Y., Hu, S., Yeo, D. J., & Cheong, K. H. (2025). Artificial Intelligence-Enabled Adaptive Learning Platforms: A Review. *Computers and Education: Artificial Intelligence*, 100429. <https://doi.org/10.1016/j.caeai.2025.100429>
- Tan, W. K., Sunar, M. S., & Goh, E. S. (2023). Analysis of the college underachievers' transformation via gamified learning experience. *Entertainment Computing*, 44, 100524. <https://doi.org/10.1016/j.entcom.2022.100524>
- Tang, L., Gu, J., Shao, M., & Zhao, L. (2025). Effects of different interventions to treat errors during STEM learning on improving young children's psychological resilience. *Research in Science & Technological Education*, 43(3), 776-795. <https://doi.org/10.1080/02635143.2024.2338809>
- Tavdgiridze, L., Didmanidze, I., KHASAIA, I., SHEROZIA, N., DOBORDGINIDZE, D., AKHVLEDIANI, D., & AKHVLEDIANI, Z. (2024). STEM teaching in contemporary education. *Challenges to National Defence in Contemporary Geopolitical Situation*, 1(1). <https://doi.org/10.3849/cndcgs.2024.296>

- Terzieva, V., Ivanova, M., Djambazova, E., & Ilchev, S. (2025). The Role of Internet of Things and Security Aspects in STEM Education. *Information*, 16(7), 533. <https://doi.org/10.3390/info16070533>
- Tran, C. T., Vuong, L. A., Nguyen, M. P., Ho, N. N. Y., & Vu, T. L. A. (2024). Factors influencing university student satisfaction with online learning in vietnam. *VNU Journal of Science: Education Research*, 40(2). <https://doi.org/10.25073/2588-1159/vnuer.4862>
- Tran, T. T. M. (2024). Vietnam's comprehensive educational reform: reflections after a decade of implementation (2013–2024). *Conhecimento & Diversidade*, 16(44), 511-537. <https://doi.org/10.18316/rcd.v16i44.12285>
- Trinh, T. M., Le, T. T. K., Le, K. M. A., Nguyen, C., & Tran, T. N. (2024). Shaping choices: factors influencing Vietnamese high school students' transition to higher education. *Higher Education*, 1-23. <https://doi.org/10.1007/s10734-024-01384-x>
- Tsabay, E., & Hewitt, D. (2025). How Assignment Deadlines and Guidelines Shape Creativity, Motivation, and Quality: Perspectives from Music-Technology Students. *Thinking Skills and Creativity*, 102073. <https://doi.org/10.1016/j.tsc.2025.102073>
- Villegas-Ch, W., Buenano-Fernandez, D., Navarro, A. M., & Mera-Navarrete, A. (2025). Adaptive intelligent tutoring systems for STEM education: analysis of the learning impact and effectiveness of personalized feedback. *Smart Learning Environments*, 12(1), 1-31. <https://doi.org/10.1186/s40561-025-00389-y>
- Vu, T., Magis-Weinberg, L., Jansen, B. R., van Atteveldt, N., Janssen, T. W., Lee, N. C., ... & Meeter, M. (2022). Motivation-achievement cycles in learning: A literature review and research agenda. *Educational Psychology Review*, 34(1), 39-71. <https://doi.org/10.1007/s10648-021-09616-7>
- Walker, K. L., Bodendorf, K., Kiesler, T., de Mattos, G., Rostom, M., & Elkordy, A. (2023). Compulsory technology adoption and adaptation in education: A looming student privacy problem. *Journal of Consumer Affairs*, 57(1), 445-478. <https://doi.org/10.1111/joca.12506>
- Wan, L., & Zhao, J. (2024). A Study on the Impact of High-Quality Content on Consumer Decision-Making from a Digital Media Perspective. In *International Conference on Frontier Computing* (pp. 360-369). Singapore: Springer Nature Singapore. https://doi.org/10.1007/978-981-96-2391-4_38
- Wang, H. H., Lin, H. S., Chen, Y. C., Pan, Y. T., & Hong, Z. R. (2021). Modelling relationships among students' inquiry-related learning activities, enjoyment of learning, and their intended choice of a future STEM career. *International Journal of Science Education*, 43(1), 157-178. <https://doi.org/10.1080/09500693.2020.1860266>

- Wang, S., Shi, G., Lu, M., Lin, R., & Yang, J. (2021). Determinants of active online learning in the smart learning environment: An empirical study with PLS-SEM. *Sustainability*, 13(17), 9923. <https://doi.org/10.3390/su13179923>
- Wei, W. (2025). Demographic Dividend. In *Dictionary of Contemporary Chinese Economics* (pp. 521-525). Singapore: Springer Nature Singapore. https://doi.org/10.1007/978-981-97-4036-9_218
- Wolf, D., & Ebner, M. (2025). Empowering Non-specialist Teachers and Students in Coding: A Case Study of a Python MOOC in an Austrian High School. In *European MOOCs Stakeholders Summit* (pp. 72-85). Cham: Springer Nature Switzerland. https://doi.org/10.1007/978-3-032-00056-9_7
- Wong, S. L., & Wong, S. L. (2021). Effects of motivational adaptive instruction on student motivation towards mathematics in a technology-enhanced learning classroom. *Contemporary Educational Technology*, 13(4), ep326. <https://doi.org/10.30935/cedtech/11199>
- Wongmahesak, K., Karim, F., & Wongchestha, N. (2025). Artificial Intelligence: A Catalyst for Sustainable Effectiveness in Compulsory Education Management. *Asian Education and Learning Review*, 3(1), 4-4. <https://doi.org/10.14456/aelr.2025.4>
- Wu, R., Gao, L., Li, J., Xie, A., & Zhang, X. (2025). Exploring key factors influencing the processual experience of visitors in metaverse museum exhibitions: an approach based on the experience economy and the SOR model. *Electronics*, 14(15), 3045. <https://doi.org/10.3390/electronics14153045>
- Wu, X., & Li, R. (2025). Perceived supports, achievement emotions, and engagement in multimodal GAI chatbot-assisted language learning: a sequential mixed-methods study. *Computer Assisted Language Learning*, 1-24. <https://doi.org/10.1080/09588221.2025.2569353>
- Xia, Q., Chiu, T. K., & Chai, C. S. (2023). The moderating effects of gender and need satisfaction on self-regulated learning through Artificial Intelligence (AI). *Education and Information Technologies*, 28(7), 8691-8713. <https://doi.org/10.1007/s10639-022-11547-x>
- Xu, Z. (2024). AI in education: Enhancing learning experiences and student outcomes. *Applied and Computational Engineering*, 51(1), 104-111. <https://doi.org/10.54254/2755-2721/51/20241187>
- Yaban, E. H., & Gaschler, R. (2025). Developmental regulation processes and age-related motivation of distance education students. *Higher Education*, 90(1), 177-197. <https://doi.org/10.1007/s10734-024-01316-9>
- Yang, H., Li, D., & Hu, P. (2024). Decoding algorithm fatigue: The role of algorithmic literacy, information cocoons, and algorithmic opacity. *Technology in Society*, 79, 102749. <https://doi.org/10.1016/j.techsoc.2024.102749>

- Yang, X. (2024). Mobile learning application characteristics and learners' continuance intentions: The role of flow experience. *Education and information technologies*, 29(2), 2259-2275. <https://doi.org/10.1007/s10639-023-11910-6>
- Yang, Y., Sun, W., Sun, D., & Salas-Pilco, S. Z. (2025). Navigating the AI-Enhanced STEM education landscape: a decade of insights, trends, and opportunities. *Research in Science & Technological Education*, 43(3), 693-717. <https://doi.org/10.1080/02635143.2024.2370764>
- Yu, J., & Tao, Y. (2025). To be in AI-integrated language classes or not to be: Academic emotion regulation, self-esteem, L2 learning experiences and growth mindsets are in focus. *British Educational Research Journal*. <https://doi.org/10.1002/berj.4180>
- Zamiri, M., & Esmaili, A. (2024). Methods and technologies for supporting knowledge sharing within learning communities: A systematic literature review. *Administrative Sciences*, 14(1), 17. <https://doi.org/10.3390/admsci14010017>
- Zardari, B. A., Hussain, Z., Arain, A. A., Rizvi, W. H., & Vighio, M. S. (2021). Development and validation of user experience-based e-learning acceptance model for sustainable higher education. *Sustainability*, 13(11), 6201. <https://doi.org/10.3390/su13116201>
- Zhai, J., Li, Z., Miao, H., Li, Z., Zhou, X., & Yang, H. (2025). Automatic Generation of Cybersecurity Teaching Cases Using Large Language Models. *Computer Applications in Engineering Education*, 33(5), e70081. <https://doi.org/10.1002/cae.70081>
- Zhai, X., & Krajcik, J. (2024). Artificial Intelligence-Based STEM. *Uses of Artificial Intelligence in STEM Education*, 1. <https://doi.org/10.1093/oso/9780198882077.003.0001>
- Zhang, A., Sun, Y., Wang, S., & Zhang, M. (2025). Research on user experience and continuous usage mechanism of digital interactive installations in museums from the perspective of distributed cognition. *Applied Sciences*, 15(15), 8558. <https://doi.org/10.3390/app15158558>
- Zhang, G., Yue, X., Ye, Y., & Peng, M. Y. P. (2021). Understanding the impact of the psychological cognitive process on student learning satisfaction: combination of the social cognitive career theory and SOR model. *Frontiers in Psychology*, 12, 712323. <https://doi.org/10.3389/fpsyg.2021.712323>
- Zhang, P., & Tur, G. (2024). A systematic review of ChatGPT use in K-12 education. *European Journal of Education*, 59(2), e12599. <https://doi.org/10.1111/ejed.12599>

- Zhao, Y., Wang, A., & Sun, Y. (2020). Technological environment, virtual experience, and MOOC continuance: A stimulus–organism–response perspective. *Computers & Education*, 144, 103721. <https://doi.org/10.1016/j.compedu.2019.103721>
- Zhong, F., & Gu, C. (2024). The impact of health information echo chambers on older adults avoidance behavior: the mediating role of information fatigue and the moderating role of trait mindfulness. *Frontiers in Psychology*, 15, 1412515. <https://doi.org/10.3389/fpsyg.2024.1412515>
- Zhou, L., Xue, S., & Li, R. (2022). Extending the Technology Acceptance Model to explore students' intention to use an online education platform at a University in China. *Sage Open*, 12(1), 21582440221085259. <https://doi.org/10.1177/21582440221085259>

Declaration of interests

- ☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- ☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: